

Damping of breathing vortices in a dipole pair-plasma condensate

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The two box-scale vortices which breathe on the zonal condensate of a strongly driven dipole pair plasma are the shear-flow instability of their own jets: in the fluid limit, jets whose shear layers are corrugated on the Debye scale are unstable in the sense of Rayleigh, and the unstable mode on the measured jets has the frequency and the radial structure of the vortices; it saturates by trapping. The vortices are damped kinetically: the field equation, with a metric that varies along the magnetic field, forces on them a small non-flute part, which is Landau damped by the particles streaming along the field, at a rate we calculate with no adjustable parameter and which is the measured one. The resonant particles carry the momentum of the damped vortices into the jets, so the jets pay for most of the loss, and the vortices drain by the small difference of two nearly cancelling exchanges with the jets, through the Reynolds stress and through the damping. The drain is slow; the collapse of the vortices in the reference simulation is set by its radial hyperdiffusion and grid.

1. Introduction

Strongly driven electron–positron plasmas in a dipole collapse, after a single explosive burst of interchange activity, into a box-scale zonal condensate, i.e., a flow that does not vary in the direction of the flow itself and holds nearly all of the energy, on which two coherent vortices ride, one on each jet, carried in opposite directions and exchanging electrostatic energy with the zonal flow at twice their advection frequency — the ‘breathing’ of Kennedy & Plunk (2026*b*). Nothing in the collisionless transit-averaged model damps the flute-like zonal part of this state (flute-like meaning constant along the magnetic field line) (Kennedy & Plunk 2026*a,b*), and the net injection of free energy by the gradients is negligible; the condensate is therefore long-lived, and the transport it carries is close to zero. The vortices, however, are not protected in the same way: in a reduced-box simulation of the same state, their energy falls slowly for several thousand time units, then ever faster, until they collapse into a weak, partly incoherent remnant. Coherent structures riding on sheared flows are common in zonally dominated plasma turbulence (Ivanov *et al.* 2020, 2025), and their lifetime can be as important as their existence. Since the vortices carry the large oscillating fluxes of the condensate, what removes them — and whether it would also remove them in a less idealised plasma — determines the long-time state of the system.

The vortices are the shear-flow instability of their own jets. The zonal flow of the condensate is not a single harmonic: its shear layers are corrugated, on a scale of a few Debye lengths, and such jets are unstable, in the sense of Lord Rayleigh (1880), to

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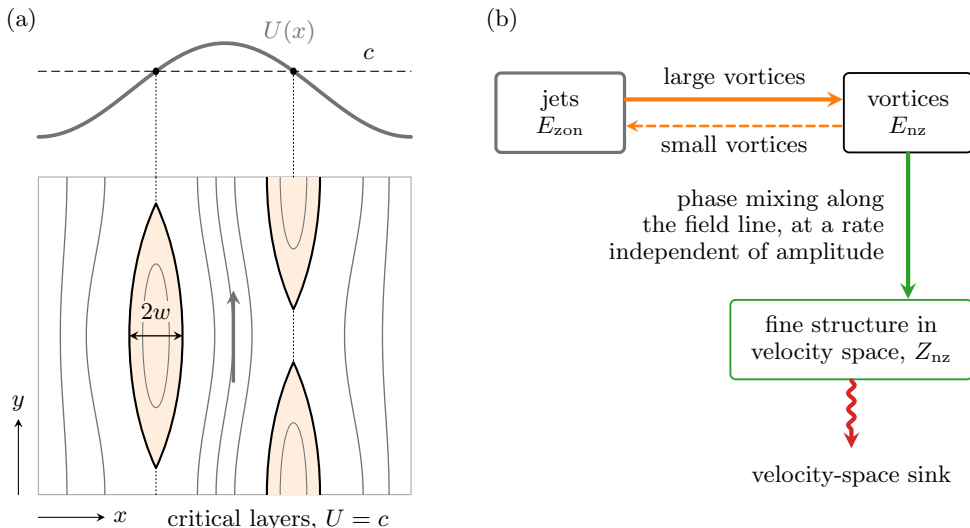


FIGURE 1. A cartoon of how the vortices lose their energy. (a) The geometry: the zonal flow $U(x)$ of one jet (grey, top) and the phase speed c of its vortex (dashed), which cross at two critical layers on the flanks of the jet (dotted); below, the streamlines in the frame of the vortex, with a cat's eye of trapped fluid, of half-width w , at each critical layer, the two rows of eyes staggered by half a wavelength, and the jet (grey arrow) meandering between them. (b) The flow of energy: while the vortices are large, the jets feed them (solid orange arrow), because the vortices are the shear instability of the jets; the electrostatic energy E_{nz} of the vortices is converted into entropy Z_{nz} , i.e., fine structure in velocity space, by phase mixing along the field line (green arrow), at a rate that does not depend on their amplitude, and removed there by the velocity-space sink (red arrow); as the vortices weaken and their jets approach stability, the exchange reverses (dashed orange arrow) and the vortices collapse. The vortices live on the instability of their jets and die of Landau damping when it fails.

perturbations of the scale of the box. The instability saturates when the fluid at its critical layers, viz., the surfaces on which the flow moves with the perturbation, is trapped in ‘cat’s eyes’ of closed streamlines (O’Neil 1965; Stewartson 1977; Warn & Warn 1978); the saturated state is the pair of breathing vortices, which, in turn, maintain the corrugation of their jets. This state is not lossless, however. A vortex is flute-like, i.e., constant along the magnetic field line, only at scales larger than the Debye length; the small part of its potential that varies along the field line does work on the particles that stream along it, and this Landau damping (Landau 1946) converts the electrostatic energy of the vortex into fine structure in velocity space, at a rate that does not depend on its amplitude. The jets make up for the loss, since the vortices draw on them, but only as long as the jets remain unstable, and the corrugation that makes them so is itself slowly eroded. The vortices and the corrugation of their jets therefore decay together, and the vortices collapse when the jets become stable. This picture is sketched in figure 1.

The damping mechanism belongs to a known family. A mode that is two-dimensional to leading order can acquire structure along the magnetic field from the geometry of the confinement, and particles that bounce or transit along the field then damp it at their orbit resonances: the rotational pumping of the diocotron mode of a non-neutral plasma, where the structure comes from the ends of the trap (Crooks & O’Neil 1995), bounce-harmonic Landau damping (Berk & Book 1969; Anderegge *et al.* 2016), and the collisionless damping of the geodesic acoustic mode of a tokamak (Sugama & Watanabe 2006). What is particular here is the origin of the structure: in a plasma that is not

quasineutral at the scale of the vortices, Poisson's equation with a metric that varies along the field line forces a non-flute part of relative size $k^2 \lambda_D^2$ on any vortex, and the vortices on which it acts are self-organised, in jets that Doppler-shift the resonance. That the particles absorb the momentum of the wave together with its energy, and that in a magnetic field the momentum becomes a radial current that drives the mean flow, is likewise known, in the theory of waves and mean flows (Andrews & McIntyre 1976; Bühler 2014), in the momentum theorems for zonal flows (Diamond *et al.* 2008), in wave-driven transport in fusion plasmas (Fisch & Rax 1992) and in the damping of diocotron and Kelvin modes (Crooks & O'Neil 1995; Kabantsev *et al.* 2014); its consequence here, that the damping takes energy from the jets and gives part of it to the vortices, is what keeps them alive.

In this paper, we first summarise the setting in §2, viz., the transit-averaged equations, the strongly driven condensate and the reduced-box simulation with its restarts. In §3, we derive the theory: the fluid limit and the instability of the jets (§3.1 and §3.2), its saturation by trapping (§3.3), the kinetic loss (§3.4) and the decay that results from the three together (§3.5). In §4, we test each of these against the simulation. §5 summarises what is established and what is open. The derivations are given step by step in appendices A to C.

2. The setting

In this section, we recall the equations of Kennedy & Plunk (2026b) and Kennedy & Plunk (2026a) to the extent needed here, describe the strongly driven condensate in which the vortices live, and specify the simulations analysed in §4.

2.1. Transit-averaged equations

We consider a symmetric electron-positron plasma, with charges $e_p = -e_e = e$, a common Maxwellian f_0 of density n_0 and temperature T , in the ordering $\rho \ll \lambda_D \ll L$, where ρ is the Larmor radius, $\lambda_D = \sqrt{\epsilon_0 T / 2n_0 e^2}$ the Debye length and L an equilibrium length. We use flux coordinates (ψ, α, l) with $\mathbf{B} = \nabla\psi \times \nabla\alpha$, and write $\delta f_a = h_a - e_a \phi f_0 / T$. Below the transit and bounce frequencies, $h_a(\psi, \alpha, \varepsilon, \lambda, t)$ is independent of l (Helander 2014; Mishchenko *et al.* 2018), where $\varepsilon = mv^2/2$ is the energy and $\lambda = v_\perp^2 / Bv^2$ the pitch-angle variable, and particles respond to the orbit-averaged potential

$$\bar{\phi}(\lambda) = \frac{1}{\hat{\tau}(\lambda)} \int_{l_1}^{l_2} \frac{\phi(l) dl}{\sqrt{1 - \lambda B(l)}}, \quad \hat{\tau}(\lambda) = \int_{l_1}^{l_2} \frac{dl}{\sqrt{1 - \lambda B(l)}}, \quad (2.1)$$

where l_1 and l_2 are the bounce points of a trapped particle or the ends of one field-line period for a passing one. To leading order in $k_\perp \rho$, the kinetic and field equations are (Abel *et al.* 2013; Mishchenko *et al.* 2018; Kennedy & Plunk 2026a)

$$\partial_t h_a + \{\bar{\phi}, h_a\} + \bar{v}_{da} \partial_\alpha h_a = \frac{e_a f_0}{T} (\partial_t \bar{\phi} - v_{*a}^T \partial_\alpha \bar{\phi}), \quad (2.2)$$

$$(1 - \lambda_D^2 \nabla_\perp^2) \frac{e\phi}{T} = \frac{1}{2n_0} \int d^3\mathbf{v} (h_p - h_e), \quad (2.3)$$

where $\{X, Y\} \equiv \partial_\psi X \partial_\alpha Y - \partial_\alpha X \partial_\psi Y$ is the $\mathbf{E} \times \mathbf{B}$ advection, $\bar{v}_{da}$ is the orbit-averaged binormal magnetic drift, which is energy dependent and of opposite sign for the two species, and $v_{*a}^T = -(T/e_a) \partial_\psi \ln f_0$ is the gradient drive at fixed energy. The dipole has no radial magnetic drift, so the zonal flow suffers no collisionless damping.

The nonlinearity of (2.2) conserves the free energy W and the electrostatic energy

E separately, and their difference $Z = W - E \geq 0$ is the entropy of the orbit-averaged perturbation (Kennedy & Plunk 2026a,b); these invariants are the closed-field-line counterparts of those of two-dimensional gyrokinetics (Plunk *et al.* 2010; Plunk & Tatsuno 2011). Here, we need only two further facts. Firstly, the free energy obeys

$$\frac{dW}{dt} = P - D, \quad (2.4)$$

where P is the injection by the gradients and $D \geq 0$ is the irreversible removal, which, in the simulations, is entirely due to hyperdiffusion (Krommes & Hu 1994; Sugama *et al.* 1996). Secondly, if we split $E = E_{\text{zon}} + E_{\text{nz}}$ into its zonal ($k_y = 0$) and non-zonal parts, the nonlinear transfer between the two satisfies

$$(\partial_t E_{\text{zon}})_{\text{NL}} = -(\partial_t E_{\text{nz}})_{\text{NL}}, \quad (2.5)$$

i.e., the zonal flow and the vortices exchange electrostatic energy without loss; this exchange is the breathing (Kennedy & Plunk 2026b). We shall write Z_{nz} and $W_{\text{nz}} = E_{\text{nz}} + Z_{\text{nz}}$ for the non-zonal parts of Z and W , and $\Gamma_E \equiv -d \ln E_{\text{nz}}/dt$ for the decay rate of the non-zonal electrostatic energy. For a structure of fixed shape, $\Gamma_E = 2\gamma_{\text{amp}}$, where γ_{amp} is the decay rate of its amplitude.

2.2. The strongly driven condensate

At $\eta \equiv L_n/L_T = 1$, the drive is above the fluid-interchange threshold (Mishchenko *et al.* 2018), and the interchange grows at the same rate at every wavelength the box allows. A single $k_y = 1$ streamer then collapses, in one burst, into $k_x = 1$ zonal bands, viz., one jet in each direction across the periodic box, modulated by two box-scale vortices, one on each jet (Kennedy & Plunk 2026b). The two vortices move in opposite directions, related by the symmetry of the pair-plasma equations under exchange of the species together with $\phi \rightarrow -\phi$ and $y \rightarrow -y$, and their relative phase sets the tilt of the flow between them, so the exchange (2.5) runs at $\omega_b = 2|\omega_{\text{adv}}|$, twice the advection frequency of either vortex. Over a breathing cycle, the state changes little; over many cycles, the vortices decay. In what follows, we call $k_y = 1$ the vortex and $k_y = 0$ the jet, and the question of this paper is why the former decays while the latter does not.

2.3. The reduced-box simulation and its restarts

All the data in this paper comes from one nonlinear simulation with the gyrokinetic code GENE (Jenko *et al.* 2000) in the dipole flux tube of Kennedy & Plunk (2026b), at $\eta = 1$, started from noise, in a box a quarter of the production box in each perpendicular direction, together with restarts of it. Its parameters are collected in table 1. There is no collision operator: all irreversible removal is by fourth-order hyperdiffusion, spectral in x and y [rate $\nu_{\perp}(\Delta x k_x/2)^4$, where Δx is the radial grid spacing, and likewise in y], and by finite-difference hyperdiffusion along the field line and in $v_{\parallel}$. One property of the grid will matter in §4.7: the radial grid spacing is comparable with the Debye length, $\lambda_D \approx 1.2 \Delta x$. The run is continued for many thousands of breathing periods after the burst, until long after the vortices have disappeared. Times are in $L_{\text{ref}}/c_{\text{ref}}$, lengths in ρ_{ref} , and energies in the units of the GENE free-energy diagnostics (Bañón Navarro *et al.* 2011, 2016).

Two sets of restarts use the same parameters. The *amplitude tests* restart the original run in the slow phase of the decay with every $k_y \geq 1$ component of the distribution function multiplied by $\epsilon = 0.3$ or 0.03 , the zonal component untouched. The *sink tests* restart it in the accelerating phase with one sink changed: the velocity-space hyperdiffusion divided by four, the perpendicular hyperdiffusion multiplied or divided

Quantity	Symbol	Value
Gradient drives	$\omega_n = \omega_T$	2.7816 ($\eta = 1$)
Grid	$n_{x0} \times n_{ky0} \times n_{z0}$	$64 \times 16 \times 32$
Velocity grid	n_v, L_v	84, 4.5
Box	$L_x, k_{y,\min}$	$3689 \rho_{\text{ref}}, 2 \times 10^{-3} \rho_{\text{ref}}^{-1}$
Debye length	λ_D	$70.7 \rho_{\text{ref}} = 1.23 \Delta x$
Perpendicular hyperdiffusion	$\nu_x = \nu_y$	2.4×10^{-3} (spectral, fourth order)
Parallel hyperdiffusion	ν_z	0.05 (fourth order)
Velocity-space hyperdiffusion	ν_v	0.2 (fourth order in $v_{\parallel}$)
Collisions	—	none

TABLE 1. A summary of the parameters of the reduced-box simulation analysed in this paper. The restarts share them, except where stated otherwise.

by four, or the radial resolution doubled. Further runs switch the radial hyperdiffusion off, or change the Debye length.

3. Theory of the vortices

In this section, we derive the three elements of the picture of figure 1, viz., the instability of the jets, its saturation and the kinetic loss, and then put them together.

3.1. The fluid limit

At the scale of the box, the jets and the vortices are flute-like, and the magnetic drift is slow compared with their motion. In this limit, the kinetic problem collapses to a fluid one, as the following argument shows; the steps are given in Appendix A.1.

Let us take the charge moment $\sum_a e_a \int d^3 \mathbf{v}$ of (2.2) with $\bar{\phi} = \phi$. The bracket commutes with the velocity integral, the gradient drive cancels between the species because $e_a (e_a f_0 / T) v_{*a}^T = -e_a \partial_\psi f_0$ sums to zero, and the drift term survives only through the even part $h_p + h_e$, since $e_a \bar{v}_{da}$ is the same for both species. Using (2.3) to write the charge moment as $(2n_0 e^2 / T)(1 - \lambda_D^2 \nabla_\perp^2) \phi$, and $\{\phi, \phi\} = 0$, we find[†]

$$\partial_t \nabla_\perp^2 \phi + \{\phi, \nabla_\perp^2 \phi\} = \frac{T}{2n_0 e \lambda_D^2} \partial_\alpha \int d^3 \mathbf{v} \bar{v}_{dp} (h_p + h_e), \quad (3.1)$$

where $\bar{v}_{dp}$ is the positron drift. The right-hand side is the effect of the magnetic drift, which is slow compared with the motion of the vortices relative to the flow of their jets; we neglect it from here on, and confirm in §4.5 that it plays no part in their energy budget. (It does not rely on the transit average either: the charge moment of the kinetic equation with the streaming of particles along the field line retained gives the same equation on averaging along the line, and the streaming adds only the loss of §3.4.) What remains is the two-dimensional Euler equation for the streamfunction ϕ and the vorticity $\nabla_\perp^2 \phi$: the Debye length drops out, the gradients drop out, as they do for a symmetric pair plasma in

[†] Strictly speaking, (3.1) holds as written in a flux tube whose metric is constant along the field line; in the dipole, $\nabla_\perp^2$ depends on l , and a flute-like ϕ satisfies it only on average along the line. This is consistent with the approximate flute-like character of the long-wavelength states (Kennedy & Plunk 2026a). In the flux tube of the simulations, the average is weighted by the Jacobian, and its effect on (3.2) is to replace k^2 in the Laplacian by $k^2 \langle g^{yy} \rangle / \langle g^{xx} \rangle$, where g^{xx} and g^{yy} are the metric coefficients; this correction is of order unity, and the stability of the jets depends on it (§4.3).

a straight field (Helander 2014), and the system of jets and vortices is an incompressible inviscid flow. Two properties of this limit matter in what follows. Firstly, multiplying the Euler equation by $-\phi$ and integrating over the box shows that it conserves $\int |\nabla_{\perp} \phi|^2 / 2$, which is the electrostatic energy E of a flute-like potential in units of $n_0 e^2 \lambda_D^2 / T$: the vortices and the jets can exchange electrostatic energy, by (2.5), but neither can lose it. Secondly, nothing in it refers to the amplitude of the gradients that drive the turbulence in which the condensate was born: the vortices are a property of the jets.

3.2. The instability of the jets

Let $U \equiv \partial_{\psi} \Phi$ be the velocity of a zonal flow with potential $\Phi(\psi)$.[†] Linearising the Euler equation about it, with $\phi = \Phi + \varphi(\psi) e^{ik\alpha - i\omega t}$, where real parts are understood, and writing $c \equiv \omega/k$ for the phase speed, gives Rayleigh's equation (Lord Rayleigh 1880; Drazin & Reid 2004),

$$(U - c) (\partial_{\psi}^2 - k^2) \varphi - U'' \varphi = 0, \quad (3.2)$$

in which U'' is the gradient of the zonal vorticity.[‡] The surfaces on which $U = c$ are the critical layers of the perturbation: there, the perturbation is at rest in the frame of the fluid, (3.2) is singular, and the perturbation exchanges energy with the mean flow. For a perturbation whose growth rate γ_R is small compared with its frequency, the exchange at the critical layers determines the growth rate (Briggs *et al.* 1970),

$$\gamma_R = \pi \frac{\sum_c U_c'' |\varphi_c|^2 / |U_c'|}{\partial_{\omega_r} \text{PV} \int d\psi k U'' |\varphi|^2 / (\omega_r - kU)}, \quad (3.3)$$

where $k > 0$, the sum runs over the critical layers, the subscript c denotes evaluation there, ω_r is the real part of the frequency, and PV denotes the principal value. Whether the perturbation takes energy from the mean flow and grows, or gives energy to it and decays, thus depends on the gradient of the zonal vorticity at its critical layers. This is the content of Rayleigh's theorem: a flow can be unstable only if U'' changes sign. The derivation of (3.2) and (3.3) and the proof of the theorem are given in appendices A.3 to A.5.

Two cases concern us. A jet that consists of a single harmonic of the box is stable to perturbations whose effective wavenumber along the flow exceeds the wavenumber of the jet, which is the case for all perturbations in the box of the simulations. A jet of the same amplitude whose shear layers are corrugated, so that each is split into strips of vorticity and U'' changes sign close to the critical layers, is unstable. The first is the zonal flow of the condensate as it would be without the vortices; the second is the zonal flow that the simulations have (§4.3).

3.3. Saturation by trapping

A growing vortex changes the trajectories of the fluid at its critical layers, and this is what stops its growth. We use local coordinates (x, y) , radial and binormal, and the $\mathbf{E} \times \mathbf{B}$ streamfunction Ψ .[¶] In the frame $\xi = y - ct$ that moves with the vortex, the flow

[†] Not to be confused with the specific flux-tube volume $U = \oint dl/B$ of Kennedy & Plunk (2026b), which does not appear in this paper.

[‡] In the flux tube of the simulations, k^2 in the Laplacian of (3.2) is replaced by $k^2 \langle g^{yy} \rangle / \langle g^{xx} \rangle$, where the angle brackets denote the average along the field line of the metric coefficients (Appendix A.2).

[¶] In GENE units, $\Psi = \phi / C_{xy}$ with C_{xy} the metric factor of the flux tube, x is in ρ_{ref} and t in $L_{\text{ref}} / c_{\text{ref}}$. GENE writes the real field as $\phi = \phi_0 + 2 \text{Re}(\phi_{k_y} e^{ik_y y})$, so the cosine amplitude of a single k_y is $A = 2|\phi_{k_y}| / C_{xy}$.

is steady, and the trajectories are the level lines of the Hamiltonian

$$\dot{x} = -\partial_\xi \mathcal{H}, \quad \dot{\xi} = \partial_x \mathcal{H}, \quad \mathcal{H} = \Psi_{\text{zon}}(x) + \Psi_v(x, \xi) - cx, \quad (3.4)$$

where Ψ_{zon} and Ψ_v are the streamfunctions of the jet and of the vortex. Let x_c be a critical layer, $U(x_c) = c$, with shear $S \equiv U'(x_c)$, and let A be the amplitude of the vortex there. Expanding (3.4) about x_c gives the Hamiltonian of a pendulum,

$$\mathcal{H} = \text{const} + \frac{S}{2} X^2 - A \cos(k\xi) + \dots, \quad X = x - x_c. \quad (3.5)$$

The fluid within a distance w of the critical layer is trapped: it circulates on closed streamlines, the ‘cat’s eye’ of Kelvin, at the frequency ω_{tr} , where

$$\omega_{\text{tr}} = |k| \sqrt{|S|A}, \quad w = 2 \sqrt{\frac{A}{|S|}}, \quad (3.6)$$

are the frequency of small oscillations about the centre of the eye and the half-width of its separatrix (Appendix B.2); both scale as $A^{1/2}$. On the two flanks of a jet, the shear has opposite signs, and the eyes are staggered by half a wavelength (see figure 1).

The growth rate (3.3) assumes that the fluid crosses the critical layer on its unperturbed trajectory. Once the fluid there has been trapped and has turned over, i.e., after a time of order $1/\omega_{\text{tr}}$, the vorticity inside the cat’s eye has been rearranged and the exchange with the mean flow ceases (O’Neil 1965; Stewartson 1977; Warn & Warn 1978). An instability that grows at the rate γ_R therefore saturates when the trapping frequency has become a fixed multiple of the growth rate,

$$\omega_{\text{tr}} = C \gamma_R, \quad \text{i.e.,} \quad A = \frac{C^2 \gamma_R^2}{k^2 |S|}, \quad (3.7)$$

where C is a pure number, which this argument does not determine (Appendix B.3). A vortex that has grown in this way has shaped its jets: at its critical layers, the fluid that it has trapped has been mixed, and the gradient of the zonal vorticity there has been flattened. The vortex thus helps to maintain the corrugation that makes its jets unstable; the two are parts of one state, and (3.7) is the relation between them. If the state changes slowly compared with ω_{tr} and γ_R , the relation is maintained: as the jets lose their instability, the vortices decay, and they disappear when the jets become stable.

3.4. The kinetic loss

A flute-like vortex in the fluid limit conserves its energy. The vortices are damped because they are not quite flute-like. The field equation (2.3) contains $\nabla_\perp^2$, whose two parts, radial and binormal, vary differently along the field line, so a potential that is constant along the line cannot satisfy it at every point: the charge of a fluid element, as it is carried round the vortex, changes by slightly different amounts at different points of the same field line. The difference is made up by currents along the line, and these require a parallel electric field, i.e., a small part φ_1 of the vortex potential that varies along the line. We show in Appendix C that

$$\varphi_1 = k^2 \lambda_D^2 \chi(l; \tilde{\omega}) \varphi_0, \quad (3.8)$$

where φ_0 is the flute-like part, $\tilde{\omega} = \omega - kU$ is the frequency of the vortex as seen by the plasma that the zonal flow carries past it, and χ is a function along the field line that the magnetic geometry fixes, through (C 11). The non-flute part is therefore small, of relative size $k^2 \lambda_D^2$, and proportional to the amplitude of the vortex.

A parallel electric field that oscillates at the frequency $\tilde{\omega}$ does work on the particles whose motion along the field line keeps step with it, viz., those whose transit or bounce frequency ω_τ satisfies $n\omega_\tau = \tilde{\omega}$ for some integer n . This is Landau damping (Landau 1946). It converts the electrostatic energy of the vortex into fine structure of the distribution in the parallel velocity, i.e., into entropy, which the sinks then remove; the rate of conversion does not depend on the sinks. Since $\tilde{\omega}$ is comparable with the transit frequency of a thermal particle over most of the width of a vortex, the resonant particles are thermal ones, and the damping of φ_1 is strong; the damping of the vortex is weak only because φ_1 is small. The result of Appendix C is

$$\frac{\mathcal{L}}{E_{\text{nz}}} = 2k^2\lambda_D^2 \frac{k^2 \int dx |\varphi_0|^2 \mathcal{D}(\omega - kU)}{\int dx (\langle g^{xx} \rangle |\partial_x \varphi_0|^2 + \langle g^{yy} \rangle k^2 |\varphi_0|^2)}, \quad (3.9)$$

where $\mathcal{L}$ is the rate at which the vortices lose electrostatic energy, the angle brackets denote the average along the field line, and the dissipation function $\mathcal{D}(\tilde{\omega}) = -\tilde{\omega} \langle s \text{Im} \chi \rangle$, with $s(l)$ the geometrical factor (C9), measures the part of φ_1 that lags behind its forcing. It vanishes for $\tilde{\omega}$ small compared with the transit frequency, which is the limit of the transit-averaged equations of §2.1, and peaks where $\tilde{\omega}$ is a few times the transit frequency.

Equation (3.9) has no adjustable parameter, and it makes three statements that can be tested. Firstly, the loss rate is independent of the amplitude of the vortices: the damping is linear, whatever the state of the trapped fluid in their cat's eyes. Secondly, it is of the order of $k^2\lambda_D^2$ times the transit frequency, so the vortices of a larger box are damped more slowly; at a fixed box it depends on the Debye length both through this factor and through the frequency of the vortices, which enters $\mathcal{D}(\tilde{\omega})$. Thirdly, the lag of the non-flute potential, measured at any radius, on either vortex and at any time, is a single function of the local value of $\tilde{\omega}$.

3.5. The decay

Let us now put these elements together. Since the Euler equation conserves the electrostatic energy, the energy E_{nz} of the vortices changes only by exchange with the jets and by the kinetic loss:

$$\frac{dE_{\text{nz}}}{dt} = \mathcal{T} - \mathcal{L}, \quad (3.10)$$

where $\mathcal{T}$ is the transfer of electrostatic energy from the jets to the vortices, averaged over the breathing, and $\mathcal{L}$ is the kinetic loss (3.9). The total electrostatic energy of jets and vortices falls only through the loss, so the jets pay for whatever part of it the vortices are resupplied. The transfer is not given by the Euler equation alone: the particles that absorb the energy of the vortices also absorb their momentum and deposit it in the jets (§4.8). Where the supply falls short of the loss, the vortices drain; as the dissipation of the simulation erodes the corrugation of the jets, the supply fails and the vortices collapse.

The theory thus makes the following predictions, which we test in §4: (i) the vortices have the frequency and the radial structure of the unstable mode of Rayleigh's equation on their jets; (ii) their trapping frequency is of the order of the growth rate of that mode; (iii) a vortex that is reduced in amplitude on unchanged jets regrows while the corrugation it needs is still in place; (iv) the vortices lose electrostatic energy to entropy through the streaming of particles along the field line, at the rate (3.9), which does not depend on their amplitude and is smaller in a larger box; (v) the jets supply the vortices with energy while they are unstable, less as they become less unstable, and finally absorb energy from them; and (vi) the slow drain of the vortices is physical, the uncompensated

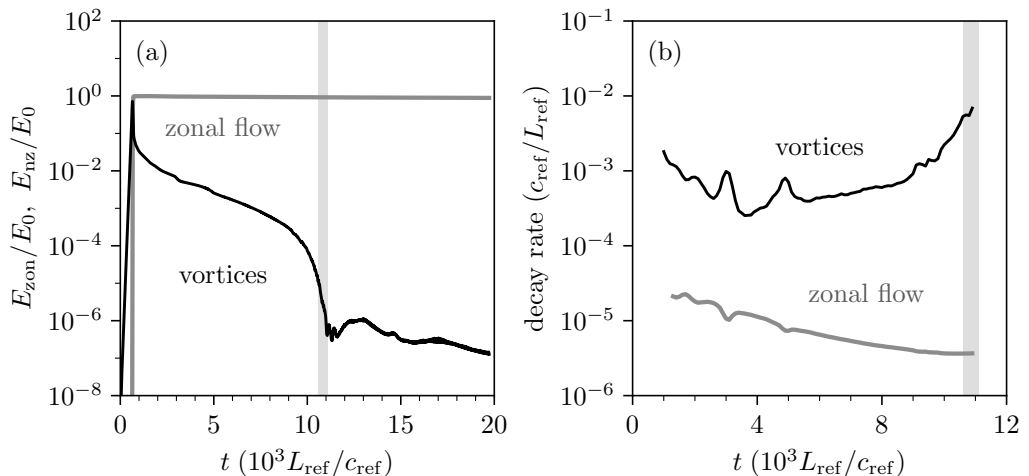


FIGURE 2. Electrostatic energy of the zonal flow and of the vortices in the reduced-box simulation. (a) The zonal energy E_{zon} (grey) and the non-zonal energy E_{nz} (black), as running means over about ten breathing periods, in units of E_0 , the largest value of E_{zon} . (b) The decay rates $\Gamma_E = -d \ln E_{\text{nz}}/dt$ (black) and $-d \ln E_{\text{zon}}/dt$ (grey), from least-squares fits of the logarithm of each energy in sliding windows of about a hundred breathing periods, from the end of the relaxation that follows the burst to the end of the collapse. The grey band marks the collapse in this and all subsequent figures. The vortices lose many decades of energy, ever faster and without a plateau, and then collapse, while the zonal flow barely changes.

part of their kinetic loss, while the dissipation that erodes the corrugation decides when they collapse.

4. Comparison with the simulations

We now test the predictions of §3.5 against the reduced-box simulation and its restarts (§2.3). All measurements of the potential are made at the outboard mid-plane, and, as in §2.2, $k_y = 1$ denotes the vortices and $k_y = 0$ the jets. The two counter-propagating vortices are separated by a two-frequency least-squares fit, $a_+ e^{-i\omega_+ t} + a_- e^{-i\omega_- t}$, in windows of about sixteen breathing periods.

4.1. The decay

The energy E_{nz} of the vortices falls by orders of magnitude between the burst and their collapse [see figure 2(a)]. The decay is not exponential. After a relaxation that follows the burst, the rate Γ_E passes through a broad minimum and then increases, ever faster, until the vortices collapse [see figure 2(b)]. Throughout, the zonal energy E_{zon} decays at a rate that is smaller than Γ_E by orders of magnitude, and the net injection P is negligible.

A vortex that is made smaller on unchanged jets does two things (see figure 3). It first regrows, relative to its rescaled initial energy, the more so the smaller it was made; it then decays faster than the original, at a rate close to that which the original run reaches when it has itself decayed to the same amplitude. The first is the instability of the jets (§4.4); the second is the dependence of the exchange with the jets on amplitude (§4.5).

4.2. The vortices and their critical layers

Each vortex propagates in the direction of its own jet, at a fixed fraction of the peak speed of the jet [see figure 4(a)]. It is radially broad — a plateau over its own jet, with

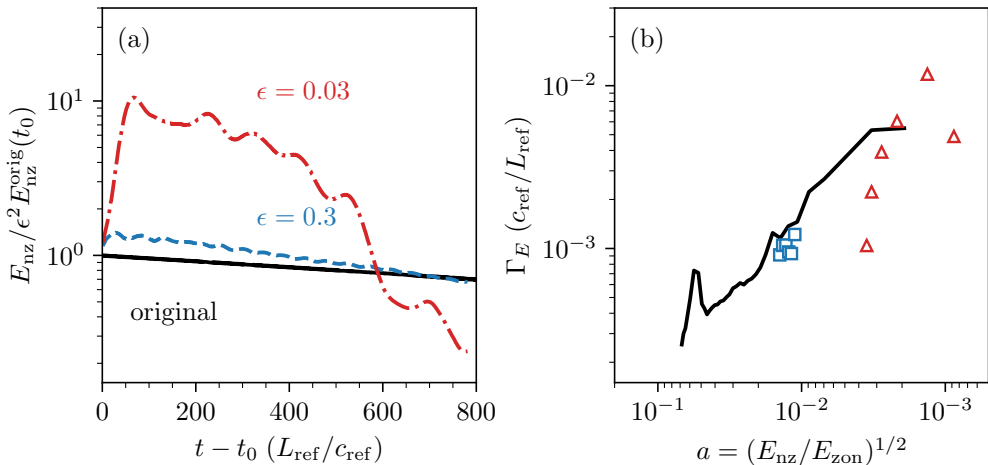


FIGURE 3. The amplitude tests. (a) The non-zonal electrostatic energy of the original run (black solid) and of the restarts with $\epsilon = 0.3$ (blue dashed) and $\epsilon = 0.03$ (red dash-dotted), divided by ϵ^2 and by the energy of the original run at the restart time t_0 , as running means over about ten breathing periods. (b) The decay rate Γ_E against the amplitude measure $a = (E_{nz}/E_{\text{zon}})^{1/2}$, which decreases to the right: the original run (black line; fits in windows of about a hundred breathing periods, from the end of the relaxation that follows the burst to the collapse) and the restarts after their regrowth (blue squares and red triangles; windows half as long). A vortex that is made smaller on unchanged jets first regrows and then decays faster than before; the weaker the vortices, the faster they decay.

maxima on both flanks — and its maxima lie just inside its critical layers, where the shear is near its largest. The streamlines of the moving-frame Hamiltonian (3.4) show the cat’s eyes directly, one on each flank of the jet, staggered by half a wavelength [see figure 4(b)].

Two premises of §3 can be checked at once. Firstly, at the wavenumber of the vortices, the drift frequency of a thermal particle† is smaller than the frequency of the vortex relative to the flow of its jet by orders of magnitude, and it displaces the resonance of a thermal particle from the critical layer by a small fraction of a grid cell, whereas the cat’s eyes are several grid cells wide. The drift is therefore negligible for the motion of the vortices. Secondly, the breathing modulates the jets by a relative amount of the order of E_{nz}/E_{zon} , which is small, so the jets may be treated as steady in (3.2).

4.3. The vortices are the unstable mode of their jets

Rayleigh’s equation (3.2), with the field-line-averaged metric of the flux tube, has one unstable mode in each direction on the measured zonal flow. Its frequency follows that of the vortices throughout the run, while both change substantially, and its radial structure is that of the vortices [see figure 5(b,c)]. This is prediction (i). The reason for the instability is visible in the zonal vorticity: each shear layer between the jets is split into two strips, and the strips sit on the critical layers of the two vortices, where the gradient of the zonal vorticity is flattened [see figure 5(a)]. This is the mark that the cat’s eyes leave on the jets, and it stays in place throughout the decay. A single harmonic of the same amplitude is stable; it is this corrugation, a few Debye lengths in scale, that

† Here the magnetic drift is the one that the code applies, viz., the curvature and ∇B drift of its geometry, averaged along the field line.

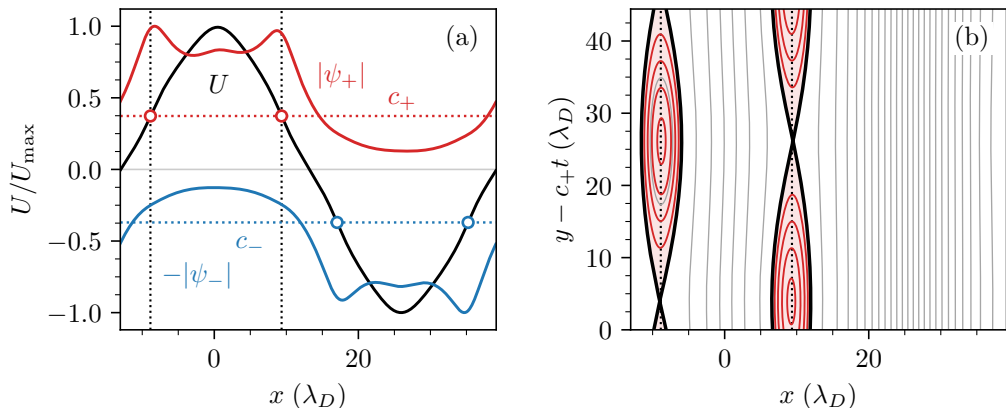


FIGURE 4. Geometry of the vortices on the zonal jets, at one time in the slow phase of the decay. (a) The zonal flow U (black), normalised to its maximum, and the radial profiles of the two vortices, $|\psi_+|$ (red) and $-|\psi_-|$ (blue), each normalised to its own maximum; the horizontal dotted lines are the phase speeds $c_{\pm}$ of the two vortices in the same normalisation, the open circles mark the critical layers $U = c_{\pm}$, and the vertical dotted lines mark those of the $+$ vortex. (b) Streamlines of the moving-frame Hamiltonian (3.4) of the $+$ vortex, with the $-$ vortex left out (grey); the separatrices are drawn in heavy black, and the trapped orbits inside them in red. The radial coordinate is shifted by a quarter of the box so that neither jet is cut by the periodic boundary. Each vortex sits on its own jet, its critical layers lie on the flanks of that jet, not at its maximum, and a cat's eye several Debye lengths wide forms at each of them.

makes the jets unstable. The growth rate falls slowly during the decay and vanishes at the collapse [see figure 5(d)].

The corrugation itself is shown in figure 6, where it is measured by the contrast of the strips, viz., the depth of the dip in the zonal vorticity between the two strips of a shear layer. The corrugation is left by the burst, changes little while the vortices are present, and decays faster once they have gone: the vortices maintain it, and without them the perpendicular sink erodes it. The jets are only just unstable. The growth rate γ_R vanishes at a finite contrast, not far below that of the slow phase of the decay, so a small loss of contrast stabilises the jets [see figure 6(b)]; this is why the growth rate, and with it the vortices, can decay to nothing while the jets and their strips hardly change.

4.4. The vortices are saturated by trapping

Figure 7 tests prediction (ii), the saturation law (3.7), with ω_{tr} measured from the vortices and γ_R computed from the jets. The law holds through the whole decay of the original run, including the collapse, with a single value of C , while the energy of the vortices falls by orders of magnitude.

The amplitude tests show how the relation is restored when it is broken. Reducing the vortices leaves them below the saturated level of their jets, and they regrow, which is prediction (iii) [see figure 3(a)]. They do not regrow to their former size, however, because, as they regrow, they flatten the corrugation of their jets further and so lower γ_R : the two approach each other. In the $\epsilon = 0.3$ restart, they meet on the curve of the original run. In the $\epsilon = 0.03$ restart, whose cat's eyes are no wider than a grid cell, the vortices stay below it [see figure 7(b)].

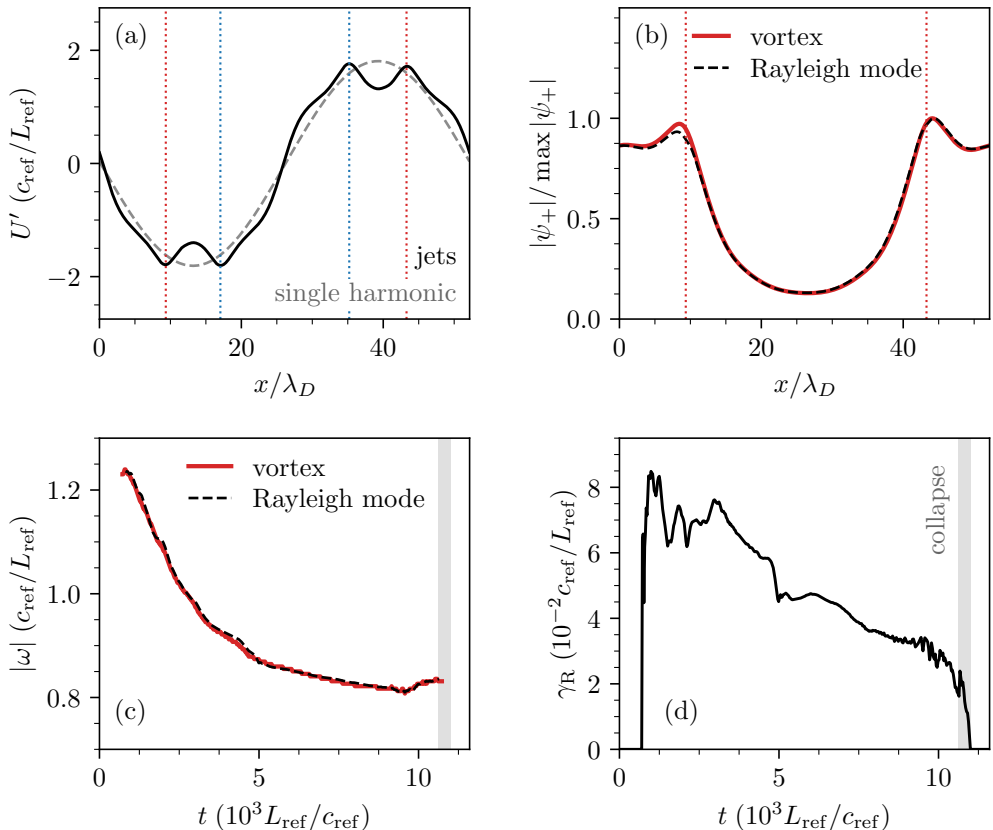


FIGURE 5. The vortices as the shear-flow instability of the jets. (a) The zonal vorticity U' of the measured jets (black) and of a single harmonic of the same amplitude (grey dashed), at one time in the slow phase of the decay; the vertical dotted lines mark the critical layers $U = \pm c$ of the vortex + (red) and - (blue). (b) The radial structure of the vortex + (red) and of the unstable mode of Rayleigh's equation (3.2) on the same jets (black dashed). (c) The frequency of the vortices (red) and of the unstable Rayleigh mode on the measured jets (black dashed) through the run. (d) The growth rate γ_R of that mode; the grey band marks the collapse. The vortices are the unstable shear mode of jets whose shear layers have split into vorticity strips, and they disappear when that instability does.

4.5. The energy budget of the vortices

The free-energy diagnostics of GENE (Bañón Navarro *et al.* 2011) resolve every term of the energy balance in (k_x, k_y) and separate the electrostatic energy from the entropy, so both terms of the budget (3.10) can be measured (see figure 8).[†]

The loss rate $\mathcal{L}/E_{\text{nz}}$ is constant while the energy of the vortices falls by orders of magnitude, and it is the same in the amplitude tests as in the original run: the loss is linear in the energy of the vortices. Nearly all of it is carried by the streaming of particles along the field line, and the small remainder by the magnetic drift.

All three statements of §3.4 about the kinetic loss hold (see figure 9). The measured

[†] All rates in this subsection are averages over the breathing, which the output resolves with only about three samples per period; the mean is fitted together with two harmonics of ω_b , and the scatter between consecutive windows gives the errors shown in figure 8. The direct action of the perpendicular sink on E_{nz} is negligible until the collapse.

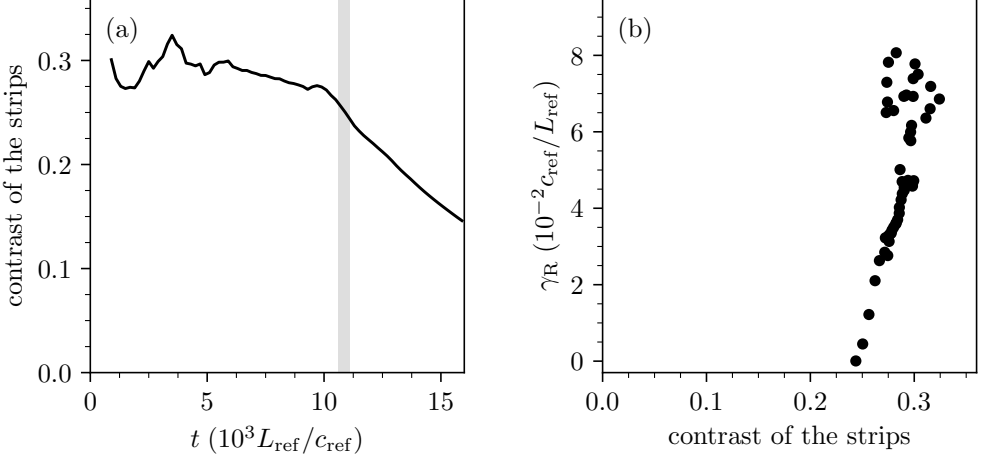


FIGURE 6. The corrugation of the jets. (a) The contrast of the vorticity strips, viz., the mean of $|U'|$ on the two strips of a shear layer less its value midway between them, divided by the largest $|U'|$ and averaged over the two shear layers, for the zonal flow averaged over about sixteen breathing periods; the grey band marks the collapse. (b) The growth rate γ_R of the unstable Rayleigh mode against the contrast, up to the collapse. The corrugation hardly changes while the vortices are present and decays once they have gone, and the jets are only just unstable: the growth rate vanishes at a finite contrast.

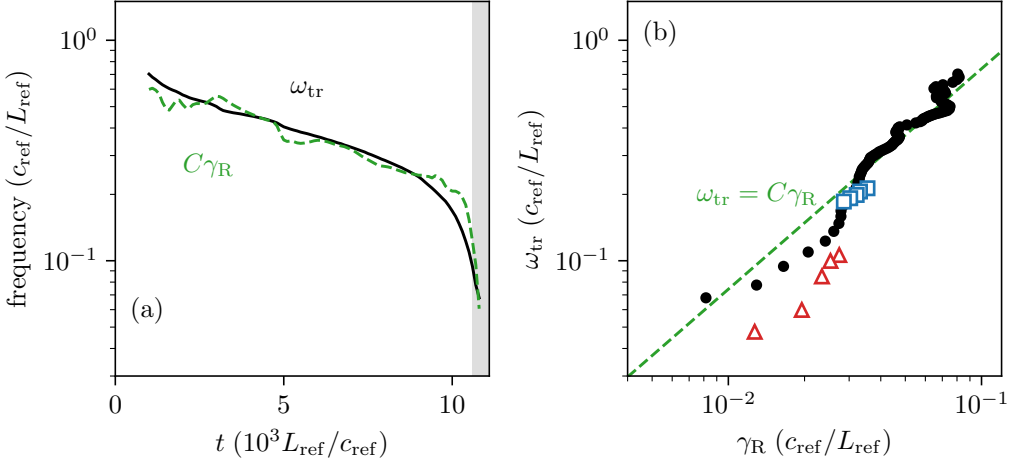


FIGURE 7. The saturation law (3.7). (a) The trapping frequency ω_{tr} of the vortices, from (3.6) at their critical layers (black), and the growth rate γ_R of the unstable Rayleigh mode of the measured jets, multiplied by the constant C (green dashed), both as means over about eighty breathing periods; C is the geometric mean of their ratio, and the grey band marks the collapse. (b) The same data, ω_{tr} against γ_R (black dots), with the line $\omega_{\text{tr}} = C\gamma_R$, and the $\epsilon = 0.3$ (blue squares) and $\epsilon = 0.03$ (red triangles) restarts, with γ_R computed on their own jets. The growth rate is that of the inviscid problem, converged in radial resolution. The trapping frequency of the vortices and the growth rate of the instability of their jets decay together, in a fixed ratio.

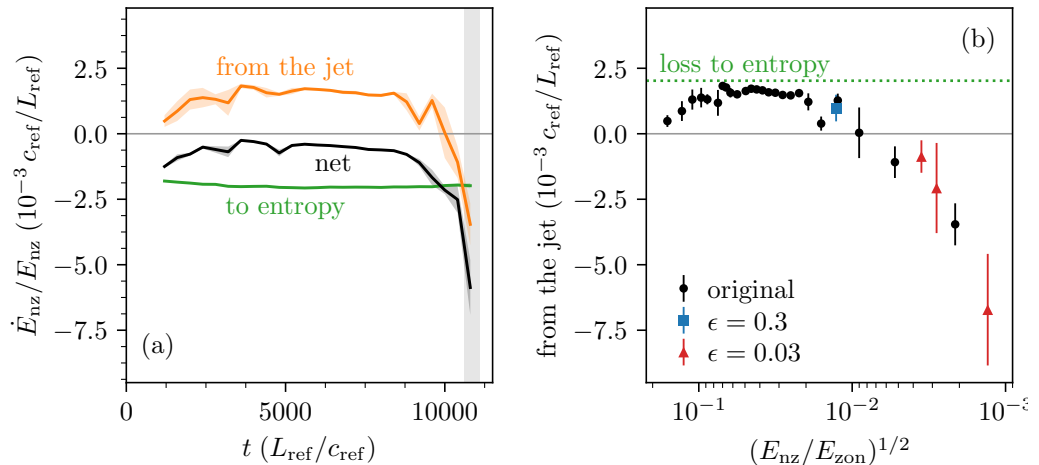


FIGURE 8. The budget (3.10) of the electrostatic energy of the vortices, averaged over the breathing. (a) The rates of change of E_{nz} , per unit E_{nz} , due to the nonlinear exchange with the zonal flow, $\mathcal{T}/E_{nz}$ (orange), due to the linear terms, which convert electrostatic energy into entropy, $-\mathcal{L}/E_{nz}$ (green), and their sum (black); the bands are standard errors over consecutive windows, and the grey band marks the collapse. (b) The exchange $\mathcal{T}/E_{nz}$ against the amplitude measure $a = (E_{nz}/E_{zon})^{1/2}$, which decreases to the right, for the original run (black dots) and for the $\epsilon = 0.3$ (blue square) and $\epsilon = 0.03$ (red triangles) restarts; the dotted line is the magnitude of the loss. The vortices lose electrostatic energy to entropy at a steady rate that does not depend on their amplitude; the jets replace most of it while they are unstable, and the decay accelerates because that supply fades and finally reverses.

lag follows the predicted $\mathcal{D}(\tilde{\omega})$, with departures close to the critical layers, where $\tilde{\omega}$ is small, and at the largest $\tilde{\omega}$. The loss rate measured by the energy diagnostics of the code, in the channel of parallel streaming, is the predicted one throughout the decay of the reduced-box run, while the energy of the vortices falls by orders of magnitude; it is the same in the restarts with reduced amplitude; and in the production run of Kennedy & Plunk (2026b), whose box is four times larger in each perpendicular direction, it is smaller by the predicted factor. In runs from noise with the Debye length doubled and halved, λ_D^2 changes by a factor of sixteen and the frequency of the vortices by a factor of eight; the measured loss rate changes by only a factor of three, and it is the predicted one in both, with no adjustable parameter. This is the remaining part of prediction (iv). The entropy that the loss produces is removed by the velocity-space hyperdiffusion, which accounts for most of the free energy removed from the non-zonal part [see figure 10(b)]; the perpendicular sink, by contrast, acts almost entirely on the zonal flow, at a rate that the vortices do not affect [see figure 10(a)].

The exchange with the jets behaves as prediction (v) requires. Throughout the slow phase of the decay, the jets supply the vortices with electrostatic energy at a rate that makes up for most of the loss, so the slow decay of figure 2 is the small difference between two larger rates. As the vortices weaken, and their jets become less unstable, the supply fades, passes through zero, and becomes a drain: the vortices give energy back to their jets, and collapse [see figure 8(b)]. The restarts follow the same curve: the $\epsilon = 0.3$ restart has the loss rate of the original run and a smaller supply, and the $\epsilon = 0.03$ restart has the loss rate of the original run and, once its regrowth is over, a drain. The acceleration of the decay, and its dependence on amplitude, are therefore those of the exchange with

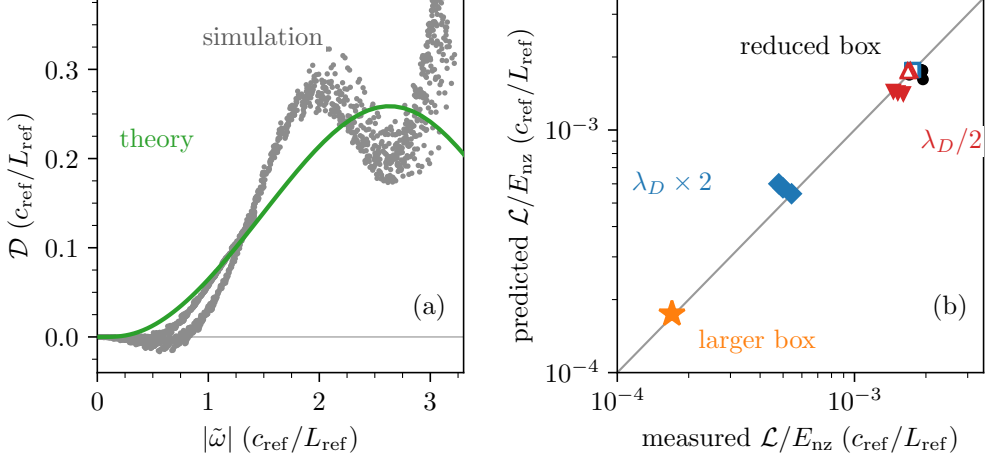


FIGURE 9. Landau damping of the non-flute part of the vortices: theory against simulation. (a) The dissipation function $\mathcal{D}$ of (C 13) against the local Doppler-shifted frequency $|\tilde{\omega}|$: the solution of (C 11) in the magnetic geometry of the flux tube (green line), and its measured counterpart, viz., the part of the non-flute potential that lags behind its forcing, at every radius where either vortex is appreciable and at many times through the decay of the reduced-box run (grey dots). (b) The loss rate $\mathcal{L}/E_{nz}$ predicted by (3.9) from the flute-like part of the vortices and the jets, against the rate measured by the energy diagnostics of the code in the channel of parallel streaming, for the reduced-box run through its decay (black dots), its $\epsilon = 0.3$ (blue squares) and $\epsilon = 0.03$ (red triangles) restarts, and the production run of Kennedy & Plunk (2026b), whose box is four times larger in each perpendicular direction (orange star); the line is equality. The vortices lose their energy by Landau damping of their non-flute part, at the rate that the theory gives, with no adjustable parameter, in every run. Diamonds and triangles: the runs with the Debye length doubled and halved.

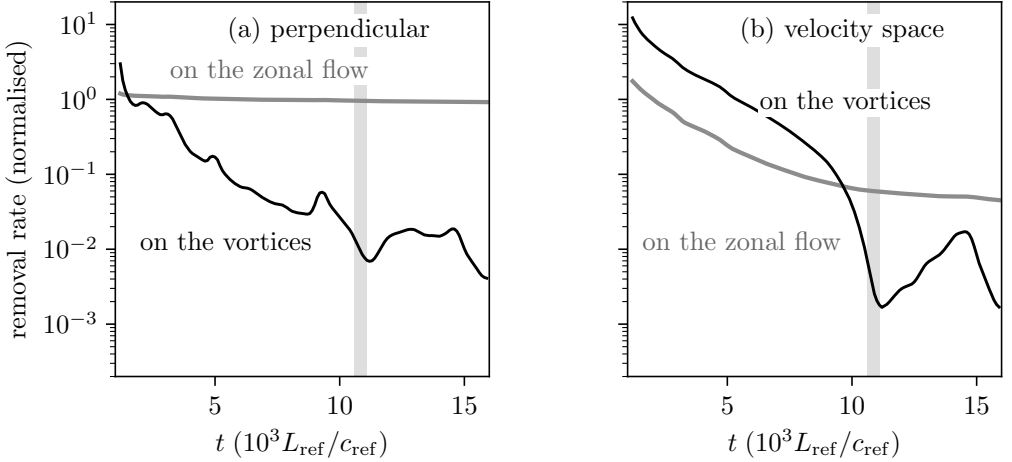


FIGURE 10. Where the free energy is removed. The rate of removal by (a) the perpendicular and (b) the velocity-space hyperdiffusion, each split into the part that acts on the zonal ($k_y = 0$; grey) and on the non-zonal ($k_y \geq 1$; black) part of the distribution, in sliding windows of about a hundred breathing periods, in units of the perpendicular removal from the zonal part in the slow phase of the decay; the parallel sink is negligible and is not shown. The grey band marks the collapse. The perpendicular sink acts on the zonal flow, whether or not there are vortices; the velocity-space sink acts mainly on the vortices, and falls away with them.

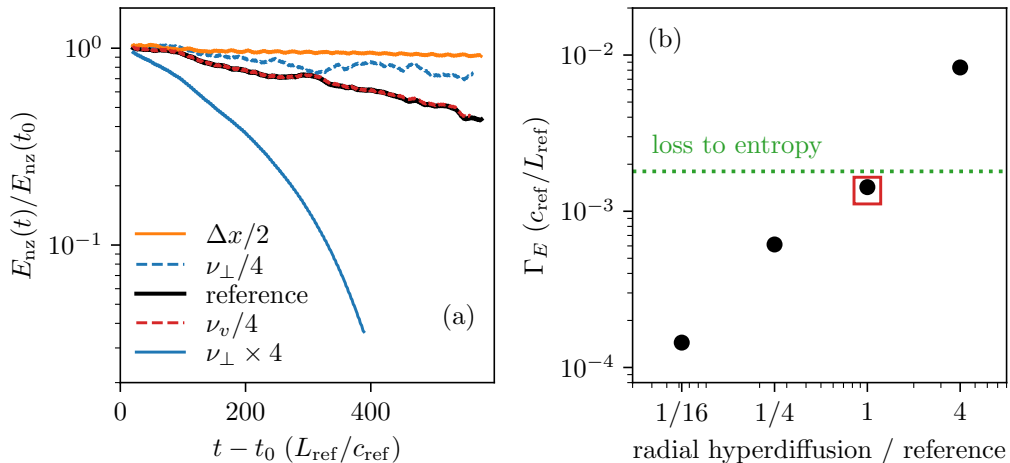


FIGURE 11. The sink tests: restarts of the reference run from one state in the accelerating phase of its decay, at time t_0 , each with one sink changed. (a) The non-zonal electrostatic energy, as running means over about ten breathing periods, for the reference run (black), for the velocity-space hyperdiffusion divided by four (red dashed, on top of the black), for the perpendicular hyperdiffusion multiplied by four (blue solid) and divided by four (blue dashed), and for the radial resolution doubled, which divides the radial hyperdiffusion at a given wavenumber by sixteen (orange). (b) The mean decay rate Γ_E over the interval of (a) against the strength of the radial hyperdiffusion relative to the reference run (black dots; the open red square is the run with the velocity-space hyperdiffusion divided by four), and the loss rate $\mathcal{L}/E_{nz}$ to entropy, which is the same in all the runs (green dotted). The decay of the vortices follows the perpendicular sink, not the velocity-space one, and where the perpendicular sink is weak the vortices persist although they are Landau damped.

the jets, not of the damping. The mean transfer from the jets is also why the zonal flow decays faster while the vortices are large than after they have gone [see figure 2(b)].

4.6. What sets the rate of the decay

Prediction (vi) is tested by the sink tests (see figure 11) and by a run repeated from noise with the radial hyperdiffusion switched off. Dividing the velocity-space hyperdiffusion by four changes neither the loss rate $\mathcal{L}/E_{nz}$, as befits Landau damping, nor the decay. Multiplying the perpendicular hyperdiffusion by four brings on the collapse at once; dividing it by four, or reducing the radial hyperdiffusion further by doubling the radial resolution, slows the late decay and removes the collapse, although the vortices go on losing energy to entropy at the same rate. Without the radial hyperdiffusion, the vortices drain exactly as in the reference run during the slow phase, and drain more slowly only later, once the reference run begins to accelerate. The slow drain is therefore physical; the acceleration and the collapse of the reference run are set by the radial hyperdiffusion and the grid. Without the radial hyperdiffusion, the vortices do not collapse, and drain at a nearly constant rate, a small fraction of their loss rate. How the loss is shared between jets and vortices depends on the resolution of the Debye scale, which sets the structure at the critical layers.

4.7. The collapse

The fluid in the cat's eyes turns over many times in the time over which the vortices decay [see figure 12(a)], which is the condition for (3.7) to be maintained as the state decays (§3.3). The decay speeds up as the cat's eyes narrow, and the vortices collapse

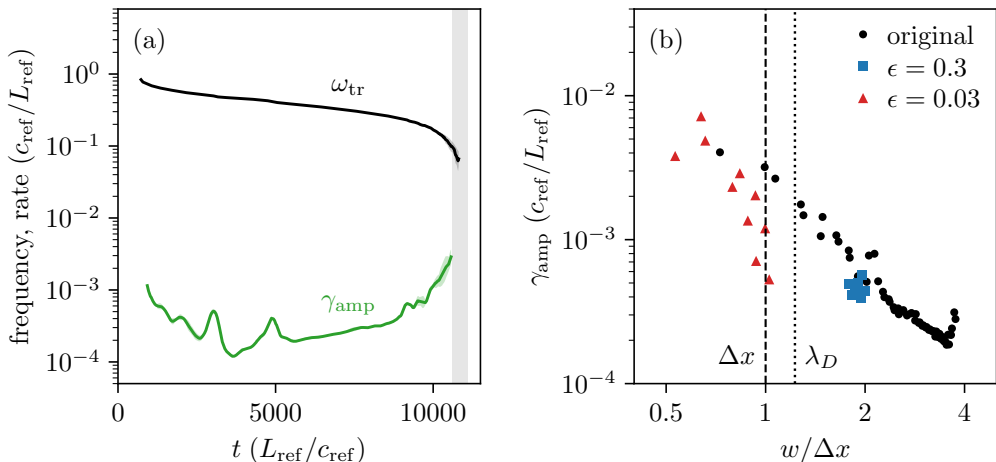


FIGURE 12. Time scales of the vortices and the approach to the collapse. (a) The trapping frequency ω_{tr} of (3.6) at the critical layer (black) and the amplitude decay rate $\gamma_{amp} = I_E/2$ (green), each as the geometric mean of the two vortices on a band that spans them; γ_{amp} is from sliding fits of the logarithm of the energy of each vortex in windows of about a hundred breathing periods, and the grey band marks the collapse. (b) The decay rate of each vortex, from fits in windows half as long, against the half-width w of its cat's eye in units of the grid spacing, for the original run (black dots) and for the $\epsilon = 0.3$ (blue squares) and $\epsilon = 0.03$ (red triangles) restarts after their regrowth; the dashed and dotted vertical lines mark $w = \Delta x$ and $w = \lambda_D$. The fluid in the cat's eyes turns over many times while the vortices decay, and the decay speeds up as the cat's eyes narrow towards one grid cell, which is also about one Debye length.

when the half-width w of the eyes has fallen to about one grid cell [see figure 12(b)], which is when the jets become stable [see figure 5(d)]. After the collapse, what is left at $k_y = 1$ is not a vortex but a weak forced response to the zonal state, supplied by the zonal flow and removed by the perpendicular sink at nearly equal rates.

In this simulation, the grid spacing is about one Debye length, $\lambda_D \approx 1.2 \Delta x$, so the scale at which the vortices cease to be flute-like and the smallest scale that the grid resolves coincide, and the collapse occurs as the cat's eyes reach that scale.

4.8. How the jets pay for the loss

The energy budget of the jets closes. In every run, the rate at which the jets lose energy is the part of the loss of the vortices that the vortices do not drain, plus the hyperdiffusion of the jets, with no adjustable constant [see figure 15(a)]. Two effects beyond the Euler equation carry this exchange, and both are measured directly from the zonal budget of the simulation.

First, the damping of the vortices is not a friction. The particles that absorb power $P(x)$ from a vortex whose Doppler-shifted frequency is $\tilde{\omega}(x)$ absorb its momentum $kP/\tilde{\omega}$ with it; in a magnetised pair plasma, the momentum displaces them radially, the electrons and the positrons in opposite directions, and the radial current changes the zonal charge, so the jets feel the force density $kP/\tilde{\omega}$ (Appendix D). The damping therefore conserves the momentum of the jets and the vortices together, while it removes the energy P . Computed from the loss formula, this force has the phase of the part of the zonal forcing that the Euler Reynolds stress does not explain, on the dominant harmonic of the jets, and it removes the energy that the jets lose beyond the Euler exchange. Where $\tilde{\omega}$ and

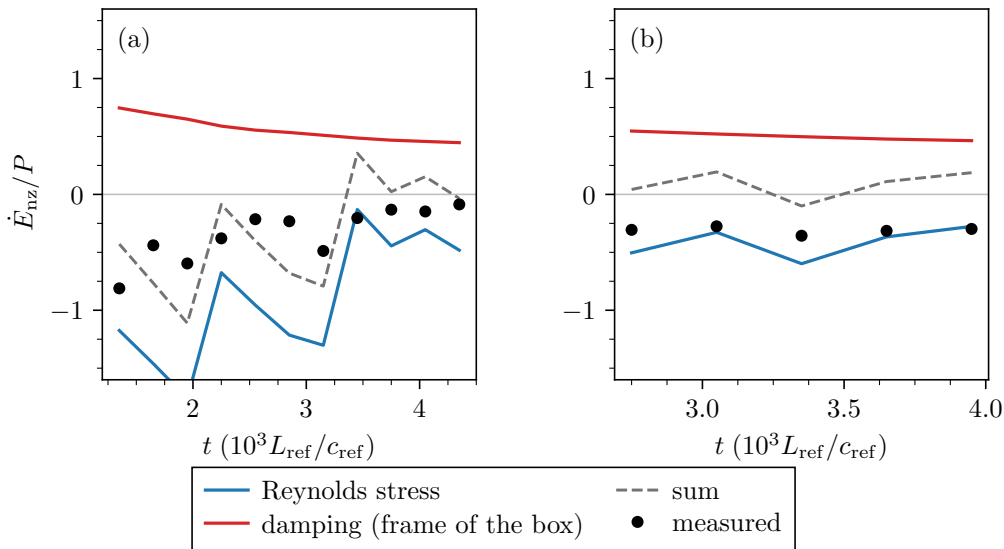


FIGURE 13. The energy of the vortices as the difference of two exchanges with the jets, in the runs without radial hyperdiffusion: (a) at the resolution of the reference run, (b) at doubled radial resolution. The rate at which the vortices gain energy through the Reynolds stress of their field-line-averaged potential (blue) and through their Landau damping in the frame of the box (red), the sum of the two (grey dashed) and the measured rate of change of their energy (dots), all divided by the absorbed power P , in windows of three hundred time units. The two exchanges are each of the order of the absorbed power and of opposite sign; the vortices drain by their small difference.

the frequency of the vortex have opposite signs, the same absorption is a gain of energy for the vortex in the frame of the box: the damping takes energy from the jets and passes part of it to the vortices.

Second, the Reynolds stress of the vortices, computed from their field-line-averaged potential, together with the deposited momentum, accounts for the nonlinear forcing of the dominant harmonic of the jets; at the fine harmonics of the jets, the Reynolds stress is the measured nonlinear forcing, and parallel streaming returns it. The two exchanges are each of the order of the power absorbed by the particles, and of opposite sign for the vortices: the Reynolds stress passes energy from the vortices to the jets, while the damping, through the momentum it deposits, passes energy from the jets to the vortices. The drain of the vortices is their small difference [see figure 13]. The near-cancellation follows from the conservation of momentum: the particles return the momentum they take from the vortices to the jets, and the Reynolds stress, which moves momentum between vortices and jets without loss, nearly restores it to the vortices; what changes the energy of the vortices is the mismatch between where the particles deposit the momentum and where the stress takes it from.

The small non-flute part of the vortices is a single function of the Doppler-shifted frequency for both vortices and at every amplitude, with a peak at the critical layers proportional to their vorticity there and independent of the radial resolution.

The exact budget of the simulation, resolved by binormal wavenumber, shows where the energy goes [see figure 14]. The jets lose energy nonlinearly at about the Landau loss rate of the vortices. Most of it reaches the vortices; the rest reaches their harmonics, the fine binormal structure of the cat's eyes, which are Landau damped as fast as they receive it, at the rate that the loss formula gives at their own wavenumber and Doppler-shifted

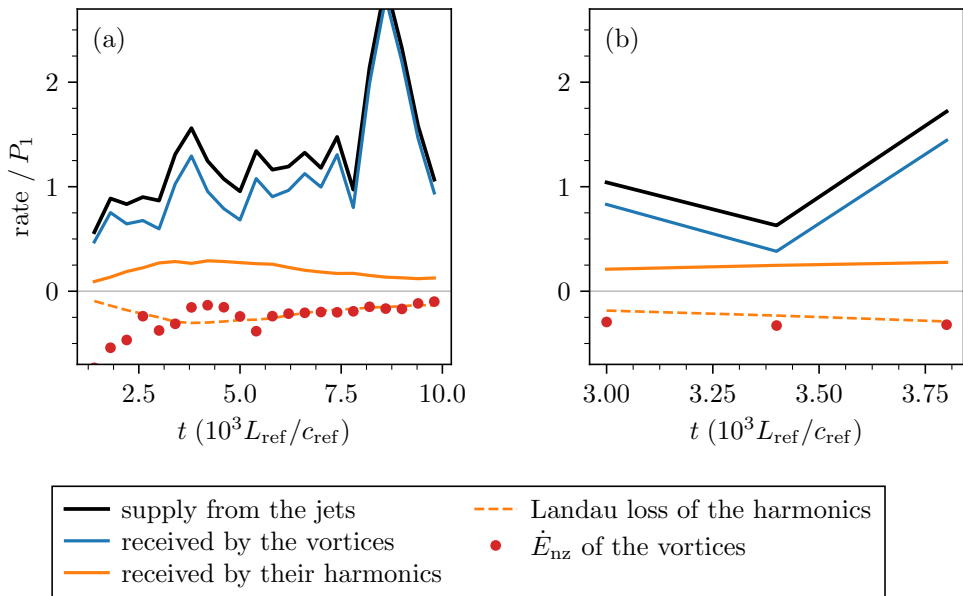


FIGURE 14. Where the energy goes, from the energy budget of the simulation resolved by binormal wavenumber, in the runs without radial hyperdiffusion: (a) at the resolution of the reference run, (b) at doubled radial resolution. All rates are divided by the Landau loss rate P_1 of the vortices: the nonlinear supply from the jets (black), the part of it received by the vortices (blue) and by their harmonics (orange), the Landau loss of the harmonics (orange dashed), and the rate of change of the energy of the vortices (red dots), in windows of eight hundred time units. The jets supply about the Landau loss of the vortices; the vortices pass a fifth of what they receive to their harmonics, which lose it at once, and drain by the remainder.

frequency. The vortices lose a little more through the magnetic drift, and drain by what is left.

5. Summary and discussion

We have given a theory of the breathing vortices of the strongly driven dipole pair-plasma condensate and tested it against a reduced-box simulation and restarts of it.

Firstly, at the scale of the box, the jets and the vortices obey the two-dimensional Euler equation, and the vortices are the unstable mode of Rayleigh's equation (3.2) on their jets: it has their frequency, throughout the run, and their radial structure. The jets are unstable because their shear layers are corrugated, on a scale of a few Debye lengths, and the instability saturates by trapping.

Secondly, the vortices are damped kinetically and linearly. The field equation forces on them a small part that varies along the field line; this part is Landau damped by the particles that stream along the line; and the resulting loss rate (3.9), which has no adjustable parameter and does not depend on the amplitude of the vortices, is the measured one, in the reduced box, in the restarts, in the larger box of the production run, and in runs with the Debye length doubled and halved.

Thirdly, the slow drain of the vortices is physical: it is the same with the radial hyperdiffusion switched off, and without it the vortices do not collapse. The jets pay for most of the loss, and the energy budget of the jets closes with no adjustable constant. Beyond the Euler equation, the damped vortices deposit their momentum in the jets

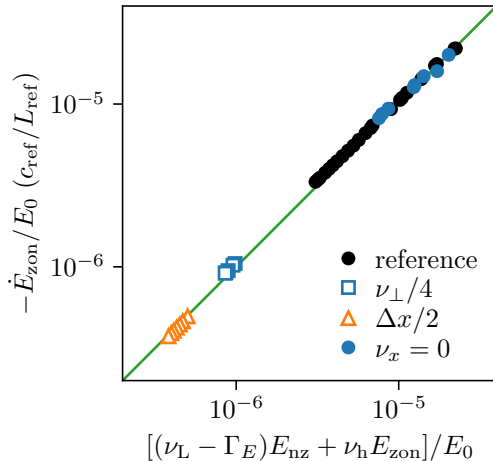


FIGURE 15. How the jets pay. The rate $-\dot{E}_{zon}$ at which the jets lose electrostatic energy, against the part of the kinetic loss of the vortices that they do not drain, $(\nu_L - \Gamma_E)E_{nz}$, plus the radial hyperdiffusion of the jets, $\nu_h E_{zon}$, in windows of about four hundred time units, for the reference run, the restarts with the perpendicular hyperdiffusion divided by four and with doubled radial resolution, and the run without radial hyperdiffusion; the line is equality and there is no adjustable constant. The jets pay for the part of the loss that the vortices do not drain.

through the resonant particles; with the Reynolds stress, this accounts for the forcing of the dominant harmonic of the jets. The two exchanges nearly cancel for the vortices, whose drain is their small difference.

Fourthly, the late acceleration of the decay and the collapse of the vortices in the reference run are set by the radial hyperdiffusion and by the resolution of the Debye scale: they follow the perpendicular sink, not the velocity-space one.

In a real plasma, the corrugation would be eroded by collisions.

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Declaration of interests

The authors report no conflict of interest.

Appendix A. From the kinetic equation to Rayleigh's equation

Here we derive, step by step, the fluid equations that govern the vortices, viz., the Euler equation (3.1) and Rayleigh's equation (3.2), and then the formula (3.3) for the growth rate of the shear-flow instability of the jets. The starting point is the transit-averaged system (2.2) and (2.3). The only assumption is that the potential is flute-like,

i.e., independent of the position l along the field line, so that $\bar{\phi} = \phi$ in (2.1); this is the case when $k_{\perp} \lambda_D \ll 1$.

A.1. The charge moment

The field equation (2.3) involves the distribution functions only through their charge moment, $\sum_a e_a \int d^3 \mathbf{v} h_a = e \int d^3 \mathbf{v} (h_p - h_e)$, so let us find the equation that this moment obeys: we multiply (2.2) by e_a , integrate over velocities and sum over the two species. Consider the terms of (2.2) in turn. Firstly, by (2.3), the charge moment itself is

$$\sum_a e_a \int d^3 \mathbf{v} h_a = \frac{2n_0 e^2}{T} (1 - \lambda_D^2 \nabla_{\perp}^2) \phi, \quad (\text{A } 1)$$

which gives the moment of $\partial_t h_a$. Secondly, a flute-like potential does not depend on the velocity, so the bracket commutes with the velocity integral, and the moment of $\{\phi, h_a\}$ is the bracket of ϕ with (A 1); since $\{\phi, \phi\} = 0$, only $-(2n_0 e^2 \lambda_D^2 / T) \{\phi, \nabla_{\perp}^2 \phi\}$ survives. Thirdly, the magnetic drifts of the two species are equal and opposite, $\bar{v}_{de} = -\bar{v}_{dp}$, and so are their charges, whence $e_a \bar{v}_{da} = e \bar{v}_{dp}$ for both, and the moment of the drift term is $e \partial_{\alpha} \int d^3 \mathbf{v} \bar{v}_{dp} (h_p + h_e)$: the drift couples the charge moment to the *sum* of the two distributions. Fourthly, on the right-hand side, $\sum_a e_a^2 \int d^3 \mathbf{v} f_0 / T = 2n_0 e^2 / T$, so the first term gives $(2n_0 e^2 / T) \partial_t \phi$. Finally, $(e_a f_0 / T) v_{*a}^T = -\partial_{\psi} f_0$ is the same for both species, so the gradient drive, multiplied by e_a and summed, vanishes. Collecting the terms, we get

$$\frac{2n_0 e^2}{T} [\partial_t \phi - \lambda_D^2 \partial_t \nabla_{\perp}^2 \phi - \lambda_D^2 \{\phi, \nabla_{\perp}^2 \phi\}] + e \partial_{\alpha} \int d^3 \mathbf{v} \bar{v}_{dp} (h_p + h_e) = \frac{2n_0 e^2}{T} \partial_t \phi. \quad (\text{A } 2)$$

The terms $\partial_t \phi$ on the two sides cancel: they describe the Boltzmann part of the response, which follows the potential adiabatically and exerts no net effect. What remains, divided by $-2n_0 e^2 \lambda_D^2 / T$, is (3.1). Three things have dropped out of it, viz., the gradients of the equilibrium, the time derivative of ϕ itself and, apart from the prefactor of the drift term, the Debye length.

The drift term on the right-hand side of (3.1) is smaller than the advection terms on its left-hand side by the ratio of the drift speed of a thermal particle to the speed of the vortex relative to the zonal flow, $|\bar{v}_{da}| / |U - c|$, which is very small for the vortices of these notes (§4.2). Without it, (3.1) is the two-dimensional Euler equation of an incompressible, inviscid fluid, written for its streamfunction ϕ : the velocity is $(-\partial_{\alpha} \phi, \partial_{\psi} \phi)$ in the (ψ, α) plane, $\nabla_{\perp}^2 \phi$ is the vorticity, and the equation states that the vorticity of each fluid element is conserved as it is carried by the flow.

A.2. The average along the field line

In a flux tube, the perpendicular Laplacian is $\nabla_{\perp}^2 = g^{xx} \partial_{\psi}^2 + 2g^{xy} \partial_{\psi} \partial_{\alpha} + g^{yy} \partial_{\alpha}^2$, where $g^{xx} = |\nabla \psi|^2$, $g^{xy} = \nabla \psi \cdot \nabla \alpha$ and $g^{yy} = |\nabla \alpha|^2$, in the normalised coordinates of the flux tube, are the metric coefficients; these vary along the field line, whereas a flute-like ϕ and the distributions h_a do not. Equation (2.3) can therefore hold only on average, and (A 1) is to be read as its average over the volume of the flux tube, $\langle \cdot \rangle \equiv \int (dl/B)(\cdot) / \int dl/B$. (The departure of ϕ from a flute-like function that restores (2.3) at every l is of relative order $k_{\perp}^2 \lambda_D^2$; it does not affect the motion of the vortices, but it is what damps them, see §3.4.) The derivation of Appendix A.1 goes through unchanged with $\nabla_{\perp}^2$ replaced by its average. In the flux tube of the simulations, $g^{xy} = 0$, so the vorticity is

$$\zeta = \langle g^{xx} \rangle \partial_{\psi}^2 \phi + \langle g^{yy} \rangle \partial_{\alpha}^2 \phi, \quad \partial_t \zeta + \{\phi, \zeta\} = 0. \quad (\text{A } 3)$$

In other words, the fluid is two-dimensional and incompressible, but its vorticity is an anisotropic Laplacian of its streamfunction.

A.3. Rayleigh's equation and the critical layer

Let the potential be that of a zonal flow, $\Phi(\psi)$, plus a small wave, $\phi = \Phi + \text{Re}[\varphi(\psi) e^{ik\alpha - i\omega t}]$, with $k > 0$, and let $U \equiv \partial_\psi \Phi$ be the velocity of the zonal flow, which is in the α direction. Any $\Phi(\psi)$ is a steady solution of (A 3), because its vorticity $\langle g^{xx} \rangle U'$ is also a function of ψ alone. To first order in the wave, the three terms of (A 3) are: the rate of change of the vorticity of the wave, $-i\omega (\langle g^{xx} \rangle \varphi'' - \langle g^{yy} \rangle k^2 \varphi)$; its advection by the zonal flow, $ikU (\langle g^{xx} \rangle \varphi'' - \langle g^{yy} \rangle k^2 \varphi)$; and the advection of the zonal vorticity by the radial velocity $-ik\varphi$ of the wave, $-ik\varphi \langle g^{xx} \rangle U''$. Their sum, divided by $ik\langle g^{xx} \rangle$, is Rayleigh's equation,

$$(U - c) (\varphi'' - k_{\text{eff}}^2 \varphi) - U'' \varphi = 0, \quad c \equiv \frac{\omega}{k}, \quad k_{\text{eff}}^2 \equiv k^2 \frac{\langle g^{yy} \rangle}{\langle g^{xx} \rangle}, \quad (\text{A } 4)$$

which is (3.2) with the metric restored; primes denote derivatives with respect to ψ . It is an eigenvalue problem: for a given jet $U(\psi)$, periodic solutions φ exist only for special values of the phase speed c , in general complex, and the wave grows at the rate $\gamma_{\text{R}} \equiv \text{Im } \omega = k \text{ Im } c$.

A *critical layer* is a radius ψ_c at which the real part of the phase speed equals the speed of the flow, $U(\psi_c) = \text{Re } c$. The fluid there moves with the wave and so feels a steady, not an oscillating, radial velocity; this is the resonance of the problem. For a neutral wave, $\text{Im } c = 0$, the coefficient of the highest derivative in (A 4) vanishes at ψ_c , and the equation is singular: by (A 4), the vorticity of the wave is $\varphi'' - k_{\text{eff}}^2 \varphi = U'' \varphi / (U - c)$, which diverges as $1/(\psi - \psi_c)$ unless $U_c'' \varphi_c = 0$, the subscript c denoting evaluation at the critical layer. For a growing wave, there is no singularity, but the vorticity is concentrated in a layer of width $\gamma_{\text{R}}/k|U_c'|$ around ψ_c . Once the wave has a finite amplitude, the fluid in this layer is trapped by it and circulates in the cat's eyes of §3.3.

A.4. When is a jet unstable?

Let us divide (A 4) by $U - c$, multiply by the complex conjugate φ^* and integrate over one period in ψ , integrating the first term by parts:

$$\int d\psi (|\varphi'|^2 + k_{\text{eff}}^2 |\varphi|^2) + \int d\psi \frac{U'' (U - c^*)}{|U - c|^2} |\varphi|^2 = 0. \quad (\text{A } 5)$$

The imaginary part of (A 5) is $\text{Im } c \int d\psi U'' |\varphi|^2 / |U - c|^2 = 0$. Therefore, a growing wave is possible only if U'' changes sign somewhere, i.e., only if the zonal vorticity U' has an extremum: this is Rayleigh's inflection-point theorem (Lord Rayleigh 1880; Drazin & Reid 2004). It is a necessary condition, not a sufficient one, as the following example, which is the relevant one here, shows.

Consider a jet that is a single harmonic, $U = U_0 \sin q\psi$. It has inflection points, but $U'' = -q^2 U$, and (A 5) becomes $\int d\psi (|\varphi'|^2 + k_{\text{eff}}^2 |\varphi|^2) = q^2 \int d\psi U (U - c^*) |\varphi|^2 / |U - c|^2$. For a growing wave, its imaginary part gives $\int d\psi U |\varphi|^2 / |U - c|^2 = 0$, and then its real part, with $U^2 = |U - c|^2 + 2U \text{Re } c - |c|^2$, gives

$$\int d\psi [|\varphi'|^2 + (k_{\text{eff}}^2 - q^2) |\varphi|^2] = -q^2 |c|^2 \int d\psi \frac{|\varphi|^2}{|U - c|^2} \leq 0. \quad (\text{A } 6)$$

The left-hand side is positive if $k_{\text{eff}} > q$. Hence a single-harmonic jet is stable to all waves whose effective wavenumber exceeds that of the jet, which is the case for the box-scale

wave in the box of the simulations. This is why the metric factor in (A 4) matters, and why a smooth jet could not, by itself, produce the vortices. The argument fails as soon as U'' is no longer proportional to U . In the simulations, each shear layer between the jets is split into two strips of vorticity [see figure 5(a)], so U'' changes sign within each shear layer, close to the critical layers, and the jets are unstable. How fast the wave then grows is the subject of the next subsection.

A.5. The growth rate

For a growing wave, (A 5) can be written as $\mathcal{D}(\omega) = 0$, where

$$\mathcal{D}(\omega) \equiv \int d\psi (|\varphi'|^2 + k_{\text{eff}}^2 |\varphi|^2) - \int d\psi \frac{kU'' |\varphi|^2}{\omega - kU}. \quad (\text{A } 7)$$

We seek its solution $\omega = \omega_r + i\gamma_R$ when the growth is slow, $\gamma_R \ll \omega_r$. The second integral is then nearly singular at the critical layers, where $\omega_r = kU$, and we need the limit of $1/(x + i\epsilon)$ as $\epsilon \rightarrow 0^+$, with $x = \omega_r - kU$. Since

$$\frac{1}{x + i\epsilon} = \frac{x}{x^2 + \epsilon^2} - i \frac{\epsilon}{x^2 + \epsilon^2}, \quad (\text{A } 8)$$

let us consider the integral of each part against a smooth function $g(x)$. The first part is odd in x and equals $1/x$ for $|x| \gg \epsilon$, while the contribution of $|x| \lesssim \epsilon$ vanishes with ϵ by symmetry; its integral thus tends to the principal value, $\text{PV} \int dx g(x)/x$, i.e., the integral with a symmetric interval around $x = 0$ excised and then shrunk to zero. The second part is a peak of width ϵ and area $\int dx \epsilon/(x^2 + \epsilon^2) = \pi$, so its integral tends to $\pi g(0)$. Therefore,

$$\frac{1}{x + i0} = \text{PV} \frac{1}{x} - i\pi \delta(x), \quad (\text{A } 9)$$

which is the Landau prescription (Landau 1946). The sign of the infinitesimal imaginary part is not a matter of choice. For a growing wave, $\text{Im } \omega > 0$ literally. More generally, the response to a disturbance applied at some instant is a superposition of terms $e^{-i\omega t}$ with ω on a line that lies above all the singularities in the complex plane, since otherwise the response would precede its cause; real frequencies must, therefore, always be approached from above (Landau 1946).

Using (A 9) in (A 7), and $\delta(\omega_r - kU) = \sum_c \delta(\psi - \psi_c)/k|U'_c|$, we find that, just above the real axis, $\mathcal{D} = \mathcal{D}_R + i\mathcal{D}_I$ with

$$\mathcal{D}_R(\omega_r) = \int d\psi (|\varphi'|^2 + k_{\text{eff}}^2 |\varphi|^2) - \text{PV} \int d\psi \frac{kU'' |\varphi|^2}{\omega_r - kU}, \quad \mathcal{D}_I = \pi \sum_c \frac{U''_c |\varphi_c|^2}{|U'_c|}, \quad (\text{A } 10)$$

where the sum is over the critical layers. For small γ_R , we expand $\mathcal{D}(\omega_r + i\gamma_R) \approx \mathcal{D}_R + i(\mathcal{D}_I + \gamma_R \partial_{\omega_r} \mathcal{D}_R)$. Here φ may be held fixed: (A 7) is the quadratic form of the operator of (A 4), which is symmetric, so it is stationary with respect to variations of φ about a solution, and the change of φ with γ_R enters only at second order. The real part of $\mathcal{D} = 0$ then determines the frequency, $\mathcal{D}_R(\omega_r) = 0$, and the imaginary part gives the growth rate, $\gamma_R = -\mathcal{D}_I/\partial_{\omega_r} \mathcal{D}_R$, i.e.,

$$\gamma_R = \pi \frac{\sum_c U''_c |\varphi_c|^2 / |U'_c|}{\partial_{\omega_r} \text{PV} \int d\psi kU'' |\varphi|^2 / (\omega_r - kU)}, \quad (\text{A } 11)$$

which is (3.3). The derivative in the denominator must be taken after the principal value. Indeed, differentiating under the integral sign would give $-\int d\psi kU'' |\varphi|^2 / (\omega_r - kU)^2$, which has a double pole at each critical layer and diverges, whereas, with $s = kU(\psi)$

as the variable of integration on each side of a layer and $G(s) \equiv U''|\varphi|^2/|U'|$, the shift $s \rightarrow s + \omega_r$ shows that

$$\partial_{\omega_r} \text{PV} \int ds \frac{G(s)}{\omega_r - s} = \text{PV} \int ds \frac{G'(s)}{\omega_r - s}, \quad (\text{A } 12)$$

which is finite for any G that is smooth at $s = \omega_r$.

Equation (A 11) has the structure of Landau's formula for the growth of a plasma wave, with the zonal vorticity gradient U'' at the critical layer in the place of the slope of the distribution function at the phase velocity of the wave. Away from the critical layers, the denominator of (A 11) is $-\int d\psi k U'' |\varphi|^2 / (\omega_r - kU)^2$, i.e., minus a weighted average of the vorticity gradient over the body of the wave. Hence the sign of γ_R : the wave grows if the vorticity gradient at its critical layers is opposite in sign to its average over the body of the wave, and decays otherwise. For a single-harmonic jet, the two have the same sign, in agreement with (A 6); on jets whose shear layers are split into strips, with the critical layers on the strips, they have opposite signs, and the wave grows. Since the fluid elements at the critical layers are trapped once ω_{tr} is comparable with γ_R , the growth stops there, which is the origin of the saturation law (3.7).

We have checked (A 11) against the growth rates obtained by solving (A 4) numerically, both for jets of the form $U \propto \sin q\psi + a \sin 3q\psi$, in which the third harmonic splits the shear layers when $a < -1/27$ and makes the jet unstable to waves with $k_{\text{eff}} > q$ once the splitting is pronounced enough, and for the jets of the simulations (§4.3). In both cases, it gives the correct sign and a growth rate close to the exact one, the difference vanishing with γ_R/ω_r , as expected of a first-order formula.

Appendix B. Trapping, the cat's eye and the saturation of the shear instability

Here we derive, step by step, the results that determine the amplitude of the vortices: that the motion of a fluid element in the frame of a vortex is Hamiltonian, (3.4); that near a critical layer it is the motion of a pendulum, (3.5), with the trapping frequency and the half-width of (3.6); and that an instability of the jets stops growing when these trapped orbits form, which gives the saturation law (3.7). Only classical mechanics and elementary fluid dynamics are needed.

B.1. Motion in the frame of the vortex

Across a strong magnetic field, every particle, whatever its charge and its energy, moves at the $\mathbf{E} \times \mathbf{B}$ velocity. In the local coordinates (x, y) , radial and binormal, this velocity is derived from the streamfunction Ψ , which is the electrostatic potential in suitable units: $\dot{x} = -\partial_y \Psi$ and $\dot{y} = \partial_x \Psi$. The flow is everywhere tangent to the contours of Ψ , since $\dot{x} \partial_x \Psi + \dot{y} \partial_y \Psi = 0$, so each fluid element would stay on one contour if Ψ did not depend on time. For a vortex that travels on a zonal flow, however,

$$\Psi = \Psi_{\text{zon}}(x) + \Psi_v(x, y - ct), \quad (\text{B } 1)$$

where the zonal part gives the flow $U(x) = \Psi'_{\text{zon}}(x)$ in the y direction, and the part due to the vortex depends on time, because the vortex moves at its phase speed c , but only through the combination $y - ct$, as long as its shape is steady. Let us, then, measure position from a point that travels with the vortex, $\xi \equiv y - ct$. Since $\partial_y = \partial_\xi$ and $\dot{\xi} = \dot{y} - c$, we have $\dot{x} = -\partial_\xi \Psi_v$ and $\dot{\xi} = U(x) + \partial_x \Psi_v - c$, whose right-hand sides are the derivatives

of a single function:

$$\dot{x} = -\partial_\xi \mathcal{H}, \quad \dot{\xi} = \partial_x \mathcal{H}, \quad \mathcal{H}(x, \xi) \equiv \Psi_{\text{zon}}(x) + \Psi_v(x, \xi) - cx. \quad (\text{B } 2)$$

These are Hamilton's equations, with ξ in the role of the coordinate and x in that of its conjugate momentum; the term $-cx$ is the streamfunction of the uniform flow $-c$ that an observer moving with the vortex sees. Since $\mathcal{H}$ does not depend on time explicitly, $d\mathcal{H}/dt = \partial_x \mathcal{H} \dot{x} + \partial_\xi \mathcal{H} \dot{\xi} = 0$: each fluid element stays on one contour of $\mathcal{H}$, viz., on a streamline of the flow as it is seen from the vortex. Equation (B 2) is (3.4); the magnetic drift, which would add a term linear in x , is negligible for the motion (§4.2).

B.2. The pendulum and the cat's eye

Without the vortex, (B 2) gives $\dot{\xi} = U(x) - c$: the fluid that flows faster than the vortex overtakes it, the fluid that flows more slowly falls behind, and the fluid on the line $x = x_c$, where $U(x_c) = c$, stays level with it. This line is the critical layer. Let us expand $\mathcal{H}$ about it. With $X \equiv x - x_c$ and the shear $S \equiv U'(x_c)$, Taylor's theorem gives $\Psi_{\text{zon}}(x) - cx = \text{const} + [U(x_c) - c]X + SX^2/2 + \dots$, in which the term linear in X vanishes by the definition of x_c . If the vortex varies little in x over the region of interest, we may also set $\Psi_v(x, \xi) \approx \Psi_v(x_c, \xi) = -A \cos k\xi$, where k is the wavenumber of the vortex, $A > 0$ is its amplitude at the critical layer, and the origin of ξ has been put at a minimum of Ψ_v . Dropping the constant, we obtain (3.5) and its equations of motion,

$$\mathcal{H} = \frac{S}{2}X^2 - A \cos k\xi, \quad \dot{\xi} = SX, \quad \dot{X} = -kA \sin k\xi. \quad (\text{B } 3)$$

The first equation of motion says that the shear carries an element at the distance X from the critical layer past the vortex at the relative speed SX ; the second, that the vortex pushes the element across the layer. Differentiating the first and substituting the second,

$$\frac{d^2(k\xi)}{dt^2} = -k^2SA \sin k\xi, \quad (\text{B } 4)$$

which is the equation of a pendulum whose angle is $k\xi$.

Let us take $S > 0$ first. The point $k\xi = 0$, $X = 0$ is the stable equilibrium of the pendulum, and is called the O-point; for small oscillations about it, $\sin k\xi \approx k\xi$ in (B 4), and the motion is harmonic with the frequency ω_{tr} given below. The points $k\xi = \pm\pi$, $X = 0$ are the unstable equilibrium, the pendulum balanced upside down, and are called X-points. The other orbits follow from the conservation of $\mathcal{H}$, whose value is $-A$ at the O-point and $+A$ at the X-points. An orbit with $\mathcal{H} < A$ cannot reach $k\xi = \pm\pi$, where (B 3) would require $SX^2/2 = \mathcal{H} - A < 0$; it goes round the O-point on a closed curve and is said to be trapped. An orbit with $\mathcal{H} > A$ has $SX^2/2 = \mathcal{H} + A \cos k\xi \geq \mathcal{H} - A > 0$, so X never changes sign: the element stays on one side of the critical layer and overtakes the vortex, or is overtaken by it, for ever. The boundary between the two families is the separatrix, $\mathcal{H} = A$, on which $SX^2/2 = A(1 + \cos k\xi) = 2A \cos^2(k\xi/2)$. Its two arcs, $X = \pm w \cos(k\xi/2)$, meet at the X-points and enclose a lens-shaped region of trapped fluid, the cat's eye [see figure 1(a) and figure 4(b)], where

$$\omega_{\text{tr}} = |k| \sqrt{|S|A}, \quad w = 2 \sqrt{\frac{A}{|S|}} \quad (\text{B } 5)$$

are the trapping frequency and the half-width of the eye at its widest, i.e., (3.6). Both scale as $A^{1/2}$. The approximations made in (B 3) hold if Ψ_v and the shear change little over the distance w .

Two further properties of these orbits will be needed. Firstly, the trapped orbits do not all have the same period: it is $2\pi/\omega_{\text{tr}}$ at the centre of the eye and increases outwards, becoming infinite on the separatrix, because a pendulum released from its inverted position takes for ever to leave it. Any pattern carried by the fluid inside the eye is therefore wound into an ever finer spiral. Secondly, if $S < 0$, the right-hand side of (B4) changes sign, so the O-point is at $k\xi = \pi$ and the X-points are at $k\xi = 0$, while (B5) holds as written. A jet has $S > 0$ on one of its flanks and $S < 0$ on the other, so a vortex whose phase is the same across the jet has its two rows of eyes displaced from each other by half a wavelength, as in figure 1(a).

When the variation of the vortex in x cannot be neglected, the same reasoning applies to the full $\mathcal{H}$. An equilibrium is a point where $\partial_x \mathcal{H} = \partial_\xi \mathcal{H} = 0$. Expanding (B2) to first order in the displacements $(\delta x, \delta \xi)$ from it gives $\delta \ddot{x} = -\mathcal{H}_{x\xi} \delta x - \mathcal{H}_{\xi\xi} \delta \xi$ and $\delta \dot{\xi} = \mathcal{H}_{xx} \delta x + \mathcal{H}_{x\xi} \delta \xi$, where the subscripts denote second derivatives at the equilibrium, and solutions proportional to $e^{\lambda t}$ exist if $\lambda^2 = \mathcal{H}_{x\xi}^2 - \mathcal{H}_{xx}\mathcal{H}_{\xi\xi}$. If this is negative, the neighbouring orbits circle the equilibrium, which is an O-point, at the frequency ω_{tr} given by $\omega_{\text{tr}}^2 = \mathcal{H}_{xx}\mathcal{H}_{\xi\xi} - \mathcal{H}_{x\xi}^2$; if it is positive, they leave it exponentially, and it is an X-point. For (B3) at its O-point, $\mathcal{H}_{xx} = S$, $\mathcal{H}_{\xi\xi} = k^2 A$ and $\mathcal{H}_{x\xi} = 0$, and (B5) is recovered.

B.3. Saturation of the instability

Suppose now that the jets are unstable, i.e., that Rayleigh's equation (3.2) has a mode that grows at the rate γ_{R} (§3.2). The linear calculation that gives γ_{R} follows each fluid element along the path that it would take without the wave, viz., $X = \text{const}$ and $\xi = \xi_0 + SXt$ by (B3), and computes the small radial displacement δX that the wave adds to it. For a wave of amplitude $A(t) \propto e^{\gamma_{\text{R}} t}$, integrating the last of (B3) along this path from the distant past gives

$$\delta X = -k \int_{-\infty}^t A(t') \sin(k\xi_0 + kSXt') dt' = -kA(t) \text{Im} \frac{e^{ik\xi}}{\gamma_{\text{R}} + ikSX}, \quad (\text{B6})$$

whose magnitude is at most $kA/(\gamma_{\text{R}}^2 + k^2 S^2 X^2)^{1/2}$. The meaning of (B6) is as follows. An element at the distance X from the critical layer sees the wave at the frequency kSX , Doppler shifted by its motion past the vortex. If $|kSX| \gg \gamma_{\text{R}}$, the push that it receives reverses many times while the wave grows by a factor of e , and its effect averages out. If $|kSX| \lesssim \gamma_{\text{R}}$, the push has the same sign for a whole growth time. It is these elements, displaced steadily across the gradient of the zonal vorticity, that pass energy from the jet to the wave [cf. (3.3)]. They occupy a resonant layer about the critical layer, of half-width $X_\gamma \equiv \gamma_{\text{R}}/|k||S|$.

The true paths, however, are those of (B3), and the unperturbed path is a good approximation to them only as long as the wave cannot turn an element round within one growth time, i.e., as long as $\omega_{\text{tr}} \ll \gamma_{\text{R}}$. By (B5), the same condition can be written as

$$\frac{w}{X_\gamma} = \frac{2\omega_{\text{tr}}}{\gamma_{\text{R}}} \ll 1, \quad (\text{B7})$$

viz., the cat's eye must be a thin sliver inside the resonant layer. As the wave grows, $\omega_{\text{tr}} \propto A^{1/2}$ grows with it, until ω_{tr} is comparable with γ_{R} . The eye has then swallowed the resonant layer, and the elements that were feeding the wave go round the eye in about the time that the wave needs to grow. An element that goes round gives the wave nothing on average: on one half of its orbit it moves across the layer one way, and on the other half it comes back. Moreover, since the orbits have different periods, the vorticity inside the eye is wound up and, after a few turns, is uniform on average along each

closed streamline, so the gradient that drove the instability has been flattened where it mattered. The growth stops. This is the argument of O’Neil (1965) for a plasma wave and the particles that resonate with it; for a shear flow, it is the theory of the nonlinear critical layer (Stewartson 1977; Warn & Warn 1978).

The wave therefore saturates at the amplitude for which

$$\omega_{\text{tr}} = C \gamma_{\text{R}}, \quad \text{i.e.,} \quad A = \frac{C^2 \gamma_{\text{R}}^2}{k^2 |S|}, \quad (\text{B } 8)$$

where C is a pure number and (B 5) has been used. This is (3.7). The argument fixes how the amplitude scales with the growth rate, the wavenumber and the shear, which is what can be tested, but not the value of C : ‘comparable’ covers anything from a fraction of a trapping period, $2\pi/\omega_{\text{tr}}$, to several periods, and the answer depends on how the vorticity is distributed near the layer. We do not derive C ; it is measured in figure 7.

Finally, let us explain why the amplitude continues to obey (B 8) while the jets change. Here γ_{R} is the growth rate on the jets as they are at a given time, so (B 8) relates two quantities that are measured from different parts of the flow, the vortices and the jets. The trapping time $1/\omega_{\text{tr}}$ and the growth time $1/\gamma_{\text{R}}$ are comparable by (B 8), and both are much shorter than the time over which the jets, and hence γ_{R} , change. If the amplitude is below the level (B 8) of the current jets, the eye is narrower than the resonant layer, and the wave grows again, at a rate of order γ_{R} , until the elements that feed it are trapped once more; this takes a few growth times, an instant on the time scale of the jets, and it is what happens when the vortices are reduced artificially [see figure 3(a)]. If, on the other hand, γ_{R} falls, the level that trapping can sustain falls with it, and the weak damping of the vortices (§4.5), which acts all the time, brings the amplitude down to that level, provided that the jets change slowly compared with ω_{tr} and γ_{R} . In other words, the amplitude follows $\gamma_{\text{R}}(t)$ adiabatically, as in figure 7(a), until γ_{R} vanishes, the jets are stable, and nothing sustains the vortices.

Appendix C. Landau damping of the non-flute part of a vortex

Here we derive the rate $\mathcal{L}$ at which a vortex loses electrostatic energy to entropy, (3.9) of §3.4. The calculation has four steps: the motion of a particle along the field line, the response of the particles to a potential that varies along the line, the part of the vortex potential that does so, and the work that this part does on the particles.

C.1. Motion along the field line

To lowest order, a particle slides along its field line with its energy $\varepsilon = mv^2/2$ and its pitch-angle variable $\lambda = v_{\perp}^2/Bv^2$ conserved, so its parallel velocity is

$$\frac{dl}{dt} = v_{\parallel} = \pm v \sqrt{1 - \lambda B(l)}. \quad (\text{C } 1)$$

If $\lambda B < 1$ everywhere, the particle is *passing*: it travels the whole length of the field line and, in the periodic flux tube, comes round again. Otherwise it is *trapped*: it is reflected where $\lambda B(l) = 1$ and bounces between two such points. Either way the motion is periodic. Its period is $\tau = \hat{\tau}(\lambda)/v$ for a passing particle and $\tau = 2\hat{\tau}(\lambda)/v$ for a trapped one, with $\hat{\tau}$ the integral defined in (2.1), and we call $\omega_{\tau} \equiv 2\pi/\tau$ the *orbit frequency*. It is proportional to v , so there are particles with every orbit frequency from zero upwards; for a thermal particle, ω_{τ} is of the order of the transit frequency, viz., the thermal speed divided by the length of the field line. We describe the position of a particle on its orbit by the angle $\theta = \omega_{\tau}t$, which increases by 2π per period, so that any function of position

along the line is, for that particle, a periodic function of θ . In particular, the potential that the particle sees is

$$\varphi(l(\theta)) = \sum_{n=-\infty}^{\infty} \varphi_n e^{in\theta}, \quad \varphi_n(\lambda) = \frac{1}{2\pi} \oint \varphi(l(\theta)) e^{-in\theta} d\theta. \quad (\text{C } 2)$$

The harmonic $n = 0$ is the orbit average $\bar{\varphi}$ of (2.1). If the potential is flute-like, i.e., the same at every point of the field line, all the other harmonics vanish. The harmonics φ_n depend on λ , which fixes the shape of the orbit, but not on the speed.

C.2. Response of the particles

Let the vortex be $\phi = \varphi(x, l) e^{iky - i\omega t}$. Its particles are carried in y by the zonal flow $U(x)$, so that, at the radius x , they see the vortex oscillate at the Doppler-shifted frequency

$$\tilde{\omega}(x) = \omega - kU(x), \quad (\text{C } 3)$$

which vanishes at the critical layers and is largest in the middle of the jet. Following a particle along its orbit, the non-adiabatic part of its distribution, $g_a e^{iky - i\omega t}$, obeys

$$\left(\frac{d}{dt} - i\tilde{\omega} \right) g_a = -i\tilde{\omega} \frac{e_a f_0}{T} \varphi(l(t)), \quad (\text{C } 4)$$

where d/dt is the rate of change along the orbit (C 1). This is (2.2) before the average over the orbit has been taken: the left-hand side is the change of g_a seen by the moving particle in the frame of the zonal flow, and the right-hand side is the rate at which the potential changes the energy of a particle, $e_a \partial\phi/\partial t$, times the slope $-f_0/T$ of the Maxwellian in energy. We have left out the magnetic drift, which is smaller than $\tilde{\omega}$ by orders of magnitude, and the gradients of the equilibrium, whose contributions to the charge are equal and opposite for the two species.

Since everything in (C 4) is periodic in θ , and $d/dt = \omega_\tau \partial/\partial\theta$, each harmonic can be solved for separately:

$$g_a = \frac{e_a f_0}{T} \sum_n \frac{\tilde{\omega}}{\tilde{\omega} - n\omega_\tau} \varphi_n e^{in\theta}. \quad (\text{C } 5)$$

The term $n = 0$ is the response to the orbit-averaged potential and is all that the transit-averaged equations of §2.1 retain. The other terms are resonant: the particles for which $n\omega_\tau = \tilde{\omega}$, i.e., those that complete n orbits in one period of the vortex as they see it, respond without limit. As in any such problem, the resonance is resolved by causality. A response that has been built up from the past is obtained by giving ω a small positive imaginary part, and then

$$\frac{1}{\tilde{\omega} - n\omega_\tau + i0} = \text{PV} \frac{1}{\tilde{\omega} - n\omega_\tau} - i\pi \delta(\tilde{\omega} - n\omega_\tau), \quad (\text{C } 6)$$

where PV denotes the principal value. To see this, replace $i0$ by $i\epsilon$: the imaginary part of $1/(X + i\epsilon)$ is $-\epsilon/(X^2 + \epsilon^2)$, a peak of width ϵ and area π , which becomes $-\pi\delta(X)$ as $\epsilon \rightarrow 0$, while the real part, $X/(X^2 + \epsilon^2)$, is $1/X$ with the neighbourhood of $X = 0$ cut out symmetrically. The first term of (C 6) is a response in phase with the potential, which exchanges no energy on average. The second is a quarter of a period out of phase: it is the resonant particles, and only they, that take energy from the potential. This is Landau damping (Landau 1946), with the orbit frequency in place of the frequency $k_\parallel v_\parallel$ of a particle streaming along a straight field.

The charge density that the two species carry between them follows by integrating

(C5) over velocities. In the units of (2.3), the right-hand side of that equation becomes

$$\frac{1}{2n_0} \int d^3\mathbf{v} (h_p - h_e) = \frac{e}{T} \rho[\varphi], \quad \rho[\varphi](l) \equiv \frac{1}{n_0} \int d^3\mathbf{v} f_0 \sum_n \frac{\tilde{\omega}}{\tilde{\omega} - n\omega_\tau + i0} \varphi_n e^{in\theta(l)}, \quad (\text{C7})$$

where $\theta(l)$ is the orbit angle at which the particle of velocity $\mathbf{v}$ passes the point l . For a flute-like potential, only $n = 0$ survives and $\rho[\varphi] = \varphi$: the particles screen a flute-like potential exactly as a Boltzmann response would, and exchange no energy with it.

C.3. The non-flute part of the vortex

A vortex cannot be exactly flute-like, because the left-hand side of (2.3) is not. In the flux tube, $\nabla_\perp^2 = g^{xx}(l) \partial_x^2 + g^{yy}(l) \partial_y^2$, where the metric coefficients g^{xx} and g^{yy} measure how the distances between neighbouring field lines change along them, and they do not vary in proportion to each other. Let us write $\varphi = \varphi_0(x) + \varphi_1(x, l)$, where φ_0 is the average of φ along the field line, weighted by the volume of the flux tube, and φ_1 is the small non-flute part. The charge moment of the kinetic equation, taken at one point of the field line, states that the charge $-\lambda_D^2 \nabla_\perp^2 \phi$ of a fluid element changes only through the divergence of the current along the line. Averaged along the line, this divergence vanishes, and what remains is the Euler equation of Appendix A: the average charge, $-\lambda_D^2 (\langle g^{xx} \rangle \partial_x^2 + \langle g^{yy} \rangle \partial_y^2) \phi$, is carried with the flow. The part of the charge that varies along the line,

$$-\lambda_D^2 [(g^{xx} - \langle g^{xx} \rangle) \partial_x^2 + (g^{yy} - \langle g^{yy} \rangle) \partial_y^2] \phi, \quad (\text{C8})$$

is carried with the same flow, and since the average charge is conserved, the rate of change of $\partial_x^2 \phi$ following the flow is $-\langle g^{yy} \rangle / \langle g^{xx} \rangle$ times that of $\partial_y^2 \phi$, which, for the vortex, is $i\tilde{\omega} k^2 \varphi_0$. Hence the charge (C8) changes at the rate $i\tilde{\omega} S(l) \varphi_0$, with

$$S(l) = k^2 \lambda_D^2 s(l), \quad s(l) \equiv \langle g^{yy} \rangle \left[\frac{g^{xx}(l)}{\langle g^{xx} \rangle} - \frac{g^{yy}(l)}{\langle g^{yy} \rangle} \right]. \quad (\text{C9})$$

A charge that varies along the line can change only through currents along the line, and only a potential that varies along the line drives such currents. The charge that φ_1 induces is, in the units of (2.3), the response (C7) less the Boltzmann part, viz., $\rho[\varphi_1] - \varphi_1$, and it changes at the rate $-i\tilde{\omega} (\rho[\varphi_1] - \varphi_1)$. Equating the two rates, we find

$$\varphi_1 - \rho[\varphi_1] = S(l) \varphi_0, \quad (\text{C10})$$

which is the part of (2.3) that varies along the line. In other words, the non-flute part of a vortex is forced by its flute-like part through the geometry of the field, in proportion to $k^2 \lambda_D^2$; it does not depend on the radial structure of the vortex, and it is linear in its amplitude. Writing $\varphi_1 = k^2 \lambda_D^2 \chi(l; \tilde{\omega}) \varphi_0$, we have

$$\chi - \rho[\chi] = s(l), \quad (\text{C11})$$

a linear equation along the field line whose only parameter is the ratio of $\tilde{\omega}$ to the transit frequency. We solve it numerically, by expanding χ in Fourier harmonics along the line and computing the orbits (C1) and the harmonics (C2) in the magnetic field of the flux tube. Because of the resonant term of (C6), χ is complex: the non-flute part of the vortex lags behind the forcing.

C.4. The loss of energy

The electrostatic energy changes at the rate $\int \phi \partial_t \varrho$, where ϱ is the charge density. The part of $\partial_t \varrho$ that is due to the motion of the particles along the field line is minus the

divergence of the parallel current, so the corresponding change of energy is minus the work done by the parallel electric field on the particles. Only φ_1 has a parallel electric field, and, by (C6), only the resonant particles absorb work on average. Carrying out the average over the orbits with (C5), we find that, per unit volume, the two species together absorb the power

$$\frac{\pi n_0 e^2}{T} \left\langle \left\langle \tilde{\omega}^2 \sum_{n \neq 0} |\varphi_{1,n}|^2 \delta(\tilde{\omega} - n\omega_\tau) \right\rangle \right\rangle = \frac{n_0 e^2}{T} \tilde{\omega} \operatorname{Im} \langle \varphi_1^* S \varphi_0 \rangle, \quad (\text{C } 12)$$

where the double angle brackets denote the average over the Maxwellian and over the flux tube, the single ones the volume-weighted average along the field line, and the equality follows by multiplying (C10) by φ_1^* , averaging, and taking the imaginary part. The left-hand side is manifestly positive: every resonance damps. The right-hand side expresses the same power through the lag of φ_1 behind its forcing. With (C11), it is $k^4 \lambda_D^4 |\varphi_0|^2 (n_0 e^2 / T) \mathcal{D}(\tilde{\omega})$, where

$$\mathcal{D}(\tilde{\omega}) \equiv -\tilde{\omega} \langle s(l) \operatorname{Im} \chi(l; \tilde{\omega}) \rangle \geq 0 \quad (\text{C } 13)$$

is a function of the Doppler-shifted frequency that the magnetic geometry fixes once and for all [see figure 9(a)]. It vanishes as $\tilde{\omega} \rightarrow 0$, where the particles average over the field line many times in a period of the vortex and the transit-averaged description applies; it is largest where $\tilde{\omega}$ is a few times the transit frequency; and it falls again at large $\tilde{\omega}$, where few particles are fast enough to resonate.

Finally, we integrate over the radius and divide by the electrostatic energy of the vortex, $(n_0 e^2 \lambda_D^2 / 2T) \int dx (\langle g^{xx} \rangle |\partial_x \varphi_0|^2 + \langle g^{yy} \rangle k^2 |\varphi_0|^2)$ per unit volume of the flux tube and unit length in y . The result is

$$\frac{\mathcal{L}}{E_{\text{nz}}} = 2 k^2 \lambda_D^2 \frac{k^2 \int dx |\varphi_0|^2 \mathcal{D}(\omega - kU)}{\int dx (\langle g^{xx} \rangle |\partial_x \varphi_0|^2 + \langle g^{yy} \rangle k^2 |\varphi_0|^2)}, \quad (\text{C } 14)$$

summed over the two vortices in the numerator and in the denominator. It contains no adjustable parameter: given the flute-like structure of the vortices and the jets, which are those of the Rayleigh problem of Appendix A, it is fixed by the geometry of the field.

Three properties of (C14) can be tested. Firstly, it is independent of the amplitude of the vortex. Secondly, it is of the order of $k^2 \lambda_D^2$ times the transit frequency, so it is small, and the smaller the longer the vortex is compared with the Debye length. Thirdly, the dissipative response, $\mathcal{D}$, is a universal function of the local Doppler-shifted frequency, on which measurements at different radii, on either vortex and at different times must all fall.

Appendix D. The momentum that the damping deposits in the jets

Here we show that the particles which absorb energy from a vortex also absorb its binormal momentum, and that they hand it to the jets. The calculation has three steps: the ratio of the energy to the momentum that a particle takes from a travelling wave, the radial displacement that the momentum produces in a strong magnetic field, and the force on the zonal flow that follows.

D.1. Energy and momentum taken from a travelling wave

Let the vortex be $\phi = \varphi(x, l) e^{iky - i\omega t}$, a pattern that travels in y at the phase speed $c = \omega/k$. In the frame that moves with the pattern, the potential does not depend on

time, so the energy of a particle measured in that frame is conserved. If, in the frame of the box, the particle gains energy $\Delta\varepsilon$ and binormal momentum Δp_y , then its energy in the frame of the pattern changes by $\Delta\varepsilon - c \Delta p_y$, and this must vanish:

$$\Delta\varepsilon = c \Delta p_y = \frac{\omega}{k} \Delta p_y. \quad (\text{D } 1)$$

This is true for any particle and any interaction with a single travelling wave; it is the reason why the energy and the momentum of a wave are absorbed together.

The power P of (C 12) is the rate at which the resonant particles gain energy in the frame of the zonal flow, which moves at $U(x)$. In that frame, the energy gained is $\Delta\varepsilon - U \Delta p_y = (c - U) \Delta p_y = (\tilde{\omega}/k) \Delta p_y$, with $\tilde{\omega} = \omega - kU$ the Doppler-shifted frequency (C 3). Hence the particles at the radius x gain binormal momentum at the rate

$$f(x) = \frac{k P(x)}{\tilde{\omega}(x)}, \quad (\text{D } 2)$$

per unit volume, and the vortex loses it at the same rate. In the frame of the box, the vortex loses energy at the rate $\omega f/k = (\omega/\tilde{\omega})P$: an amount P goes into the random motion of the particles, which is the loss of §3.4, and the rest, Uf , goes with the momentum.

D.2. From momentum to a radial current

In a strong magnetic field, the binormal momentum of a guiding centre is tied to its radial position. The canonical momentum conjugate to y is $p_y = mv_y + q_a A_y$, and in the slab coordinates of the flux tube $A_y = Bx$ to the order needed here; the kinetic part mv_y is negligible for the slow motions considered. A particle of charge q_a that gains binormal momentum Δp_y is therefore displaced radially by

$$\Delta x = \frac{\Delta p_y}{q_a B}. \quad (\text{D } 3)$$

The same result follows from the drift of a particle under a force F_y in a magnetic field, $v_x = F_y/(q_a B)$. The resonant electrons and positrons both gain momentum in the same direction, so they are displaced in opposite directions, and together they carry the radial current

$$J_x = \sum_a q_a n_a \frac{\Delta x_a}{\Delta t} = \frac{f}{B}, \quad (\text{D } 4)$$

whatever the share of the momentum taken by each species.

D.3. The force on the jets

A radial current that varies with x changes the charge density, $\partial_t \rho = -\partial_x J_x = -\partial_x f/B$. In the flute-like zonal state, the charge is the polarisation charge of the zonal flow, which, in the units of (2.3), is proportional to the zonal vorticity $\partial_x U$, with U the $\mathbf{E} \times \mathbf{B}$ velocity. The change of charge is therefore a change of the zonal flow,

$$\frac{\partial U}{\partial t} = f, \quad (\text{D } 5)$$

in the fluid units of §3.1: the momentum that the particles take from the vortex is given to the jets. The jets gain energy at the rate $\int U f dx$, which is the remainder of the energy lost by the vortex in the frame of the box; the particles keep $\int P dx$ as heat. The momentum of the jets and the vortices together is thus conserved by the damping, while their energy decreases by the absorbed power.

Where $\tilde{\omega}$ has the sign of ω , i.e. between the critical layers of a vortex, f has the sign of the momentum of the vortex, and the damping takes energy from the vortex. Where $\tilde{\omega}$ and ω have opposite signs, in the cores of the jets that move faster than the vortex, the same absorption gives the vortex energy in the frame of the box, $-(\omega/\tilde{\omega})P > 0$, and takes it from the jets. Since $\mathcal{D}(\tilde{\omega})$ grows with $|\tilde{\omega}|$ (figure 9), most of the absorption takes place in the cores of the jets, and the net effect of the damping on the energy of the vortices, in the frame of the box, is a gain.

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